ANNEX 1

Horizon Europe (HORIZON)

Description of the action (DoA)

Part A

Part B

DESCRIPTION OF THE ACTION (PART A)

Associated with document Ref. Ares(2022)7691164 - 08/11/2022

COVER PAGE

Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.

PROJECT *Grant Preparation (General Information screen) — Enter the info.*	
Project number:	101091934
Project name:	Electrochemical synthesis of hydrogen peroxide from water, air and renewable electric energy
Project acronym:	Power2Hype
Call:	HORIZON-CL4-2022-TWIN-TRANSITION-01
Topic:	HORIZON-CL4-2022-TWIN-TRANSITION-01-15
Type of action:	HORIZON-RIA
Service:	HADEA/B/03
Project starting date:	fixed date: 1 January 2023
Project duration:	48 months

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PROJECT SUMMARY

Associated with document Ref. Ares(2022)7691164 - 08/11/2022

Project summary

Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about.

Use the project summary from your proposal.

In line with the European Green Deal, the Processes4Planet partnership and the European initiative for more affordable, secure and sustainable energy (REPowerEU), the RIA proposal Power2Hype aims to establish a sustainable route for the hydrogen peroxide production. Based on air and water as feedstocks, and renewable energy as the sole energy source, the POWER2HYPE route addresses an economically viable and green alternative to the classical energydemanding anthraquinone oxidation (AO) process. Hydrogen peroxide will be produced by paired electrolysis at both the cathode and anode using an innovative custom-made 200% electrolyser. Aiming the electrification and decentralization of the chemical industry, the POWER2HYPE concept proposes the TRL 6 demonstration of electrolytic hydrogen peroxide production at any given concentration (from 20 to 99 %) at the site where it is needed. With this flexibility of concentration range, hydrogen peroxide could be used from sanitation, bleaching to niche markets as propulsion fuels.

LIST OF PARTICIPANTS

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
1	COO	FHG	FRAUNHOFER GESELLSCHAFT ZUR FORDERUNG DER ANGEWANDTEN FORSCHUNG EV	DE	999984059
2	BEN	WU	WAGENINGEN UNIVERSITY	NL	999981634
3	BEN	WR	STICHTING WAGENINGEN RESEARCH	NL	999547365
4	BEN	FAU	FRIEDRICH-ALEXANDER-UNIVERSITAET ERLANGEN-NUERNBERG	DE	999995408
5	BEN	DIA	DIACCON	DE	923552230
6	BEN	EI	ENERGIEINSTITUT AN DER JOHANNES KEPLER UNIVERSITAT LINZ VEREIN	AT	983246321
7	BEN	SOLVAY	SOLVAY SA	BE	998067921
7.1	AE	RHDO	RHODIA OPERATIONS	FR	998153863
7.2	AE	SSPI	SOLVAY SPECIALTY POLYMERS ITALY SPA	IT	999932455
7.3	AE	SRLF	Rhodia Laboratoire du Futur	FR	998394811
8	BEN	TRSA	TECNICAS REUNIDAS SA	ES	996847176
9	BEN	SGL	SGL FUEL CELL COMPONENTS GMBH	DE	885436856
10	BEN	SLVG	SOLVGE B.V.	NL	890326917
11	BEN	LOBA	GLOBAZ, S.A.	PT	983211983
12	BEN	IRIS	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES	998993786

LIST OF WORK PACKAGES

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
WP1	Coordination	1 - FHG		1	48	D1.1 – Manual of management D1.2 – Set-up and operation of a secured project data management folder D1.3 – POWER2HYPE risk register D1.4 – Annual Reports including risk analysis
WP2	Advanced Materials	9 - SGL		1	24	D2.1 – Report on gas diffusion layer characterization and stability D2.2 – Membrane availability for demonstrator on the aromatic bipolar membranes proposed solution D2.3 – Development of economically viable BDD electrodes for the electrolytic H2O2 production D2.4 – Catalyst formulation and the role of polarity and porosity on carbon D2.5 – Report on the development of gas diffusion electrodes for H2O2 production

WP3	Development of convergent paired electrolysis of H2O2	3 - WR		1	30	D3.1 – Report on Highly conductive, chemically and thermally stable AEMs and CEMs D3.2 – Report on cathodic half-cell development and continuous cathodic H2O2 production D3.3 – Continuous production of H2O2 by convergent paired electrolysis

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
WP4	Cell design and process integration	5 - DIA		4	30	D4.1 – Report on the design for pilot electrolyser D4.2 – Definition of the layout for integrated electrolysis process available D4.3 – Report on upscaling & integration of concentrator
WP5	Pilot plant design, build and commission	8 - TRSA		16	36	D5.1 – Report on basic plant design including: i) developed process design; ii) equipment specification; iii) preliminary operating manual D5.2 – Delivery of the final pilot electrolyser ready to be integrated D5.3 – Pilot plant commissioning, manual of operation and testing campaign report D5.4 – Process monitoring and control with a digital twin

WP6	Demonstration	7 - SOLVAY		30	48	D6.1 – HAZOP analysis report D6.2 – Report on 1000h continuous operation D6.3 – Report on operation flexibility
WP7	Technology impact assessment	6 - EI		6	48	D7.1 – Definition and specification of evaluation scenarios of commercial deployment D7.2 – Report on techno-economic assessment and site analysis D7.3 – Report on GHG emissions and complete Life Cycle Assessment KPIs D7.4 – Policy brief on socio-technical

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						and regulatory barriers and recommended adjustments
WP8	Dissemination, communication and exploitation	11 - LOBA		1	48	D8.1 – Dissemination & communication plan activities and updates D8.2 – Data Management plan D8.3 – IPR and Innovation Data base D8.4 – Exploitation plan report and updates (incl. business plans)

Work package WP1 – Coordination

Work Package Number	WP1	**Lead Beneficiary**	1. FHG
Work Package Name	Coordination		
Start Month	1	**End Month**	48

Objectives

To coordinate and supervise all scientific activities to be carried out;
To carry out the overall administrative and financial management of the project;
To manage the contract with the European Commission and the Consortium Agreement;
To manage and monitor project progresses and ensure that the project reaches the objectives in time (deliverables, milestones, etc.);
To perform quality control and risk management;
To sustain and facilitate regular communication internally and with the European Commission.

Description

Task 1.1: Scientific coordination (FHG) M1-48
Strategic project management will be done by the project coordinator (PC) in consultation with the General Assembly (GA, the highest decision-making body in POWER2HYPE with representatives of all partners). This will include the review of technical reports, the overall project implementation and the achievement of project objectives and critical milestones. The project coordinator will ensure that project progress matches work plan and timeline. Should deviations occur, FHG will put special actions into place to get the work back on track and/or take the decision to the GA if necessary. The PC will follow up actions related to decisions made by the GA. Regular scientific exchanges will be planned and conducted with support of EI and TRSA.

Task 1.2: Administrative, financial coordination and reporting (FHG, All) M1-48
FHG will be the day-to-day contact point for partners, implementing best practices, explaining rules, expectations, roles, procedures, etc. FHG will provide proactive support, consolidate project-level data, and ensure efficient communication channels. A document repository, accessible through the data sharing platform, will provide the partners with a tool to store and share documents, to arrange meetings and to follow up actions. Procedures and tools for administrative, contractual and day-to-day coordination shall be set-up at the project start via the Manual of Management. FHG will outline the roles and responsibilities of the different partners in detail (Project Coordinator, General Assembly, Management Team, etc.). FHG will also set-up a consortium agreement to be signed before the signature of the grant agreement regulating the internal organisation mechanisms and IPR issues. FHG shall regularly monitor progress towards deliverables and milestones and remind partners three month in advance when deliverables and reports are due. For periodic reports, FHG will consolidate this information from WP leaders and produce reports in accordance with the requirements of the EC Grant Agreement. Furthermore, FHG will manage the administration and timely distribution of EU funds, calculation of budget adjustments and/or budget shifts between partners. Amendments will be initiated and submitted by the coordinator where necessary. Consortium meetings will be planned every 6 months. FHG will act as an interface between all project beneficiaries and the EC Officers.

Task 1.3: Risk management and contingency planning (FHG, All) M1-48
The regular reporting, follow-up & communication detailed will enable the coordinator and the FHG management team to closely monitor the quality of project results with respect to defined key performance indicators (KPI). In case of doubt regarding the quality of specific results, an external evaluation can be performed. A risk register will be continuously maintained by FHG and will record eventual issues challenging the achievement of project deliverables and milestones. Risks will be assessed regarding severity and likelihood of occurrence as high, medium or low. As soon as risks are identified and the severity is ranked, appropriate mitigation strategies will be identified, agreed, and implemented. The impact of these mitigation measures will be monitored closely.

Work package WP2 – Advanced Materials

Work Package Number	WP2	**Lead Beneficiary**	9. SGL

Work Package Name	Advanced Materials		
Start Month	1	**End Month**	24

Objectives

The development of advanced, industrially viable materials and cell components (catalysts, gas diffusion layers/ electrodes, ion-exchange membranes and planar BDD anodes) for large-scale electrochemical production of H2O2. All materials are required to:
withstand the oxidative environment generated by concentrated H2O2
provide high selectivity towards electrochemical conversion of air and water to H2O2 be
compatible with electrochemical filter press-type reactor designs enable faradaic
efficiencies of 160 % (ORR+WOR) at current densities of 400 mA cm2.

Description

Task 2.1: Development of robust gas diffusion layers (GDL) and efficient gas diffusion electrodes (SGL, WU, FAU, RHDO) M1-24

SGL will initially screen different GDL candidates (carbon fiber paper-based GDL with varied density, hydrophobicity levels, microporous layer properties) with respect to gas transport, electrolyte barrier properties and long-term stability against concentrated H2O2. Liquid barrier properties will be investigated by means of contact angle and bubble point measurements also usable for the evaluation of stability of GDLS and GDEs against concentrated H2O2.
After identification of the most suitable GDL support, gas diffusion electrodes (GDE) will be manufactured by sheet (A4 size) or roll-to-roll (R2R) coating of the GDL (roll width 520 mm) with catalyst inks based on different metals, carbides or carbons (provided by WU/Task 2.2 or commercially available reference materials) and ionomers (provided by RHDO Task 2.4). SGL will optimize the inks with respect to rheology and ionomer to catalyst ratio (I:C) and will explore effects of catalyst layer thickness and conductive additives to improve the electrode performance.
In parallel, FAU performs BDD-coating on the electrode side of the support material in a 2-step process: (i) CVD infiltration with boron-doped titanium nitride at 1000 °C to suppress sp2-carbon diffusion during step 2; (ii): CVD-BDD coating to suppress cathodic H2 generation.

Task 2.2: Synthesis and characterization of carbon and carbide based catalyst materials (WU, FAU, WR) M1-24

Synthesis, characterization and fast electrochemical screening by rotating ring disk electrode (RRDE) of noble metalfree catalysts will be carried out by WU. Novel carbon based atomically dispersed non-noble metal electro-catalysts in the presence and absence of a dopant will be developed for enhancing selectivity to 2e− ORR through multiple active sites while maintaining high activity. The following key actions will be performed:
i) Synthesis of carbon supports: mesoporous carbon and carbon nanofibers with high surface area will be synthesized.Mesoporous carbon (CMK-3) will be synthesized at WR. Mesoporous carbon nanofiber by WU.
ii) Deposition of atomically dispersed non-noble and abundant metals on mesoporous carbon supports: WU will developnon-noble single metal catalyst based on mesoporous carbon supports. Copper single atom on mesoporous carbon supports for ORR will be evaluated and will be compared to commercial palladium counterparts. Further, the effect of nitrogen doping in mesoporous carbon material will be investigated in combination with single atom metal species. RRDE measurements will determine the electrocatalytic activity of all carbon materials toward ORR to H2O2 in acidic, neural and in alkaline conditions.
iii) Catalysts characterization: carbon catalysts will be characterized before and after screening by BET, zeta potential,STEM-HAADF, XPS to get insight on porosity, surface charge, particle size and electronic properties.
iv) Synthesis of carbides on BDD surface reaction of carbide forming metals (Ti, Cr, W): by combining inhomogeneousshort time PVD sputtering of the metals and subsequent annealing between 400°C -800°C for carbide forming with the diamond layer, mixed surface island with carbidic and non-noble metallic shares will be evaluated by FAU.

Task 2.3: Development of low-cost boron-doped diamond (BDD) electrodes (FAU, FHG) M1-24

BDD-niobium anodes 10 cm x 10 cm will be produced by FAU with different boron content and diamond grain size to enhance the WOR efficiency. To replace the cost-intensive niobium, low-cost BDD substrates will be developed by BDD coating of electrically isolating porcelain or cordierite. By changing the phase content of the ceramic mass, a coefficient of thermal expansion (CTE) near the CTE of CVD diamond should be generated to minimize the thermal stresses in the BDD. An important target of this task is the improvement of the ampacity of the BDD-ceramic by introducing metallic interlayers (titanium) for better electrical conductivity. BDD sample will be delivered to FHG for testing in WP3.

Task 2.4: Development of innovative ion exchange ionomers for membrane and cathode binder (RHOD, SGL, WU) M1-24

SSPI will develop further its polymer backbones to design new family of ionomers based on aromatic polymer to provide proton and anion exchange ionomers with high chemical resistance and durability under “harsh” conditions. Electrolyte containing concentrated H2O2 will be used to simulate ageing conditions and the most promising aromatic ionomer candidates will be screened. Key parameters to consider are the aromatic backbone selection, Mw, charge carrier nature and EW versus ageing tests. Three different aromatic backbones will be investigated: (i) aromatic proton exchange membrane formulation, conductivity and stability tests; (ii) aromatic anion exchange membrane formulation, conductivity and stability tests and aromatic ionomer as binder for the catalyst (WU) on the GDL cathode (for Task1/ SGL); (iii) bipolar membrane configuration: preparation/formulation strategies (planar versus bulk heterojunction) to

maximize surface contacts for CEM/AEM domain to improve H2O2 generation, the electrochemical performance will be evaluated by FHG and WR in WP3.

Work package WP3 – Development of convergent paired electrolysis of H2O2

Work Package Number	WP3	**Lead Beneficiary**	3. WR
Work Package Name	Development of convergent paired electrolysis of H2O2		
Start Month	1	**End Month**	30

Objectives

To enhance the performance indicators by improving both half-cell reactions utilizing the materials developed in WP2.
To achieve a current density ≥400 mA/cm2, H2O2 concentrations ≥10% (cathode) & ≥3% (anodic) and energy usage of ≤4 kWh/kg 100 wt.% H2O2, which implies Faraday efficiencies of at least 80%.
To integrate the anodic and cathodic half cells at an energy usage of ≤3 kWh/kg 100 wt.% H2O2.

Description

Task 3.1: Membrane and ionomer optimization (SSPI, WR, FHG) M1-18

SSPI will optimize the formulation of the Aquivion® proton exchange membrane (PEM), Aquivion® anion exchange membrane (AEM) and ionomer to increase conductivity and improve stability towards concentrated H2O2. The membranes performance will be evaluated based on their electrical conductivity, water transport, chemical stability and perms electivity. SSPI will compare the performance of the optimized Aquivion® PEM and AEM with membranes from WP2. The selection of the most suitable membrane composition will be made in conjunction with WR and FHG. SSPI will produce selected PEM and AEM for testing in following tasks of WP3 (T3.2, T3.3, T3.4) and will supply upscaled sizes for WP4, WP5 and WP6.

Task 3.2: Cathodic half-cell development (WR, WU, SGL, RHDO, FHG) M6-24

MEA fabrication strategies will be evaluated by WR in close cooperation with RHDO, SGL and WU. WR will investigate the manufacture conditions and the influence of the materials composition (WU catalysts, GDL from SGL, AEM and ionomer from SSPI). A half-cell will be designed by WR in conjunction with FHG in which the performance MEA can be evaluated. WR together with SSPI will evaluate porous solid polymer electrolyte (PSPE) strategies in the light of design, fabrication method and cell integration. WR will use mathematical modelling to evaluate PSPE strategies on electrical and hydrodynamic resistance. The most promising PSPE strategies will be fabricated by WR and SSPI, and the electrical resistance will be evaluated with impedance spectroscopy. Optimized PSPE will be used for electrolysis its performance on the reduction of the cell voltage will be analyzed. The stability and performance indicators of the cathodic halfcell with MEA and PSPE will compared with established OER (from RHDO). WR will develop a processing system enabling continuous operation under controlled conditions to determine the stability of the fabricated cathodic half-cell for 200 hours.

Task 3.3: Anodic half-cell development (FHG, FAU, WR) M6 -24

FHG will optimize the reaction conditions utilizing the BDD anodes (WP2) for the anodic water oxidation reaction (WOR). To progress towards paired electrolysis reaction, conditions such as pH, ionic species and concentrations in the anolyte will be evaluated. Evaluation of BDD samples and reaction conditions will be done in a parallel screening setup consisting of 6 divided cells. The optimal conditions and outperforming BDD anodes will be used in a custom designed half-cell focusing on low energy consumption. FHG will design this half-cell in conjunction with WR as similarly described in T3.2. After optimizing reaction conditions, the nature of the anode and design of the anodic half-cell will be essential to determine the stability and performance indicators. FHG will demonstrate continuous WOR to H2O2 and stable operation under controlled condition for 200 hours.

Task 3.4: Towards convergent paired electrolysis (FHG, WR, SSPI) M12-30

FHG and WR will devise strategies to effectively couple the anodic and cathodic reactions using compatible electrolytes, as well as matching ion and water flows. If needed, the design of the half-cell (and subcomponents like the catalyst) will be modified to better fit a paired electrolyser. FHG together with WR will, iteratively, optimize the conditions for the paired electrolysis. Final step is the demonstration of the paired electrolyser under continuous and stable operation for 200 hours, also testing performance under load-flexible operation.

Work package WP4 – Cell design and process integration

Work Package Number	WP4	**Lead Beneficiary**	5. DIA
Work Package Name	Cell design and process integration		
Start Month	4	**End Month**	30

Objectives

To design custom-made electrolyser for the paired anodic and cathodic production of H2O2 at 400 mA/cm2 and 3 kWh/kg H2O2 with approx. 0.2 m2 total electrode area (200 - 400 cm2 per electrode)
To develop stable cell components (monopolar and bipolar plates)
To simulate the physical parameters of the cell (fluid dynamic and heat transfer)
To scale up and integrate a DSP device for concentrating H2O2.

Description

Task 4.1: Cell design and lab electrolyser fabrication (DIA, FHG, WR) M4-24

These sub-tasks will be composed of three main steps:
i) Definition of suitable electrolyser setups regarding low resistance/high efficiency/costs: design study regarding theelectrolyser setup (bipolar/monopolar) and the implementation of novel components & materials.
ii) Design/construction/production of lab-scale electrolysers (approx. 10 cm2): DIA will produce two lab electrolysersfor WR and FHG. Solutions for GDE-, membrane-, electrode-fixation and sealing of all the components will be analyzed. The electrolysers must perform with low resistance and high efficiency setup.
iii) Design of demonstrator electrolyser (approx. 0.2 m2 total electrode area, 400 mA/cm2): based on feedback fromWP3 lab-cells, DIA will design the final pilot electrolyser. Special considerations are on the stack-able system and a later serial production (costs, handling and availability). The final design will be transferred to WP5.

Task 4.2: Fluid dynamic and heat transfer simulations multiphysics (TRSA, FHG) M6-24

The innovative design of a paired electrochemical production of H2O2 requires innovative flow path designs and stack architecture. These will be carried out by TRSA aided by Computational Fluid Dynamic (CFD) and multiphysics modelling tools. First, information on the utilized materials, structural designs and expected flows in the cell system from subtask 4.1.1 will be acquired. Then, sets of flow paths will be designed and simulated to assess their virtual performance. FHG will contribute with heat evolution (ohmic losses) and dissipation simulations. Interaction between simulated designs and testing of preliminary prototyped cells in subtask 4.1.2 will take place until optimal designs are fabricated in subtask 4.1.3.

Task 4.3: Corrosion assessment (DIA) M4-18

All materials used for cell construction will be tested by DIA for corrosion resistance in the specific electrolytes (acidic and alkaline) of the half cells. H2O2 will be tested as oxidation agent. Tests will be carried out in a static support for non-conducting materials and under electrochemical load for materials that will conduct electricity.

Task 4.4: Process integration (FHG, DIA, WR), M18-30

Based on the target performance parameters (FE = 160% at J > 400 mA/cm2 and 3 kWh/kgH2O2), the results of WP3 and the design of the prototype electrolyser, FHG will develop suitable gas and fluid flow (e.g. single pass / multi pass / feed-and-bleed) and operation parameters (e.g. stack voltage and current) for the paired electrolysis. The result of this

task will be a process layout for the combined H2O2 production and a mass flow diagram that will be the basis for the pilot system in WP5.

Task 4.5: Integration of eH2O2 with concentration-device (SLVG) M4-30

Based on the H2O2 electrosynthesis, the technical requirements for to the quality of H2O2 will be set envisioning potential product applications. SLVG will development the concentration device further and scale it up to produce 20 kg H2O2/day. A compatible interface between the electrolytic H2O2 production and DSP will be developed to enable a continuous concentration process. The energy efficiency of the concentrator will be improved by optimization of energy consumption and performances under different H2O2 concentrations. The concentrator will be transferred to TRSA for integration with the electrolyser into one single pilot unit (WP5).

Work package WP5 – Pilot plant design, build and commission

Work Package Number	WP5	**Lead Beneficiary**	8. TRSA
Work Package Name	Pilot plant design, build and commission		
Start Month	16	**End Month**	36

Objectives

To design, build, and test a pilot plant to produce H2O2 using only air, water and electricity with a capacity of up to 20 kg/day of H2O2 (dry basis) diluted in water at an outlet concentration of up to 99%
To commission an integrated and fully operating prototype plant to be demonstrated in WP6.

Description

Task 5.1. Construction of the stack (M16-28) (Leader: DIA; Participants: TRSA, FHG)

This task will receive as input a design of the electrolyser stack from WP4, including key cell performance parameters like temperature, FE, J, targeted H2O2 concentration, electrolyte nature, crossover and mass flow rates. T5-1 will be mainly dedicated to stack construction by DIA for its further integration in the pilot plant by TRSA. The stack will bear a total cell surface of approx. 400 cm2 and a current density of 400 mA/cm2 according to preliminary calculations to eventually yield up to 20 kg H2O2/day. DIA and FHG will realize a campaign of thorough tests to ensure full functionality of the stack prior to its integration into the plant.

Task 5.2. Design of the pilot plant (M16-30) (Leader: TRSA; Participants: DIA, SOLVAY, SLVG, IRIS, FHG)

This task provides the basis conceptual design for the pilot plant, including operation modes and regimes (renewable energies transients, shutdown, max-min turnover rates, etc). T 5.1 is harmonised with T7.1 on the definition of representative reference cases for up-scale from kW- to MW-scale in implementation scenarios. T5.2 includes as activities for the design of the plant, but not limited to, basic engineering, heat and mass balance (H&MB), piping and instrumentation diagrams (P&IDs), process flow diagrams (PFDs), safety systems, heat management, electrolytes management, by-products separation (O2, H2), power electronics, safety assessment (HAZOP), and H2O2 concentration management.

Task 5.3. Digitalization of the pilot plant (M16-36) (Leader: IRIS; Participants: TRSA)

This task will be divided in two main actions:
i) Development of monitoring and control system. The key requirements for the monitoring and control of the pilot plantcontinuous process will be studied, considering: system structure, production objectives/constraints, process coupling, product routing variables, controlled/manipulated variables and production control strategy. TRSA will design the control loops and IRIS will provide support on this. IRIS will provide the monitoring, control and safety systems architecture. Sensors and control systems will be selected and connected to a process monitoring software platform

based on a Patbox system (IRIS) to acquire, store and to apply, if needed, model treatment for statistical process monitoring of the data coming from sensors.
ii) Development of digital twin linked to case study. This task will develop a highly sophisticated Digital Twin (DT)combining innovative mechanistic modelling with advanced real-time data analytics. The DT will implement a hybridbased strategy combining both statistical and data-driven models with mechanistically derived equations enabling robust on-line predictions of key process performance indicators. Appropriate AI techniques (e.g. Machine Learning, Deep Learning…) will be defined for building models using the experimental process data from the monitoring system. The objective is to act as a virtual decision support tool to adjust the process parameters for process improvement through the models developed and real data gathered from the processes.

Task 5.4. Construction of the pilot plant (M22-30) (Leader: TRSA; Participants: DIA, SLVG, IRIS).

This includes as activities, but not limited to, the procurement of materials, construction of the rack, installing the electrochemical stacks provided by DIA, installing the concentration unit provided by SLVG, and construction of the balance of plant with the set-up of the monitoring and control system provided by IRIS. All the activities involved in the construction will follow standard specifications given by the technology developers and corresponding safety procedures. The pilot plant will be fit in a container to facilitate mobility and transport to demo sites as well as to demonstrate decentralized production of H2O2.

Task 5.5. Commissioning and testing of the pilot plant (M31-36) (Leader: TRSA; Participants: DIA, SLVG, SOLVAY)

TRSA will commission the pilot plant and run a test campaign to confirm and optimize operability for demonstration in WP6. Eventually, the plant in container will also be set up and shipped to demo site. For this shipping set-up, it will be essential to receive from WP4 and apply basic safety and maintenance measures, such as, but not limited to, stack membrane dry-out prevention.

Work package WP6 – Demonstration

Work Package Number	WP6	**Lead Beneficiary**	7. SOLVAY
Work Package Name	Demonstration		
Start Month	30	**End Month**	48

Objectives

The objective is to run a medium-scale to evaluate feasibility, time, cost and scaling factors to properly design and implement a full-scale commercial electrolytic hydrogen peroxide production process in order to prove the new technology using commercially available, pre-tested equipment. The production rate has a capacity of up to 20 kg/day of H2O2 (dry basis) diluted in water at an outlet concentration of up to 99%.

Description

Task 6.1. Safety assessment (M30-38) (Leader: SOLVAY, Participants: TRSA, SLVG)

Solvay will perform a complete evaluation of the safety risks being assumed prior to test or operation of the system, design, and procedural hazards that may be present in the system being acquired, and procedural controls and precautions that should be followed to operate it safely, considering the final demo location. HAZOP methodology assessment.
TRSA and SLVG will participate to provide insights on the components of the demonstration unit.

Task 6.2. Process operation continuously for 1000h. Plant testing, data monitoring and analysis (M36-44) (Leader: SOLVAY, Participants: TRSA)

This task includes the commissioning of the Demo unit at Solvay site, operation and safety training to the operator. Afterwards, the unit will run continuously for 1000 hours on steady-state operation in order to test the robustness of the

system, allowing to estimate the different components lifetime under industrial operating conditions (i.e. current density of 400 mA/cm2 and 20 kg H2O2/day).

Task 6.3. Validation upon intermittent electrical grid loading (SOLVAY, EI, IRIS) M44-48

In order to provide data for LCA & TEA in WP 7, the unit will run to simulate different electrical grid loading (taking advantage of periods where the peak demand is lower). It is estimated to run at 50% capacity during electricity peak demand, and 100% capacity during electricity low demand. A comparison with virtual demonstration will be carried out by IRIS.
Finally, the produced H2O2 will be validated in terms of applicability in industrial processes in collaboration with industrial end-users as external partners (i.e. DOW). For that purpose, samples of produced H2O2 will be tested by the external partner in different reactions.

Work package WP7 – Technology impact assessment

Work Package Number	WP7	**Lead Beneficiary**	6. EI
Work Package Name	Technology impact assessment		
Start Month	6	**End Month**	48

Objectives

To model process design upscale and implementation scenarios
To assess environmental footprint of the electrochemical H2O2 production through Life Cycle Assessment
To assess potential production cost for H2O2 and their development in relevant application scenarios, including technology learning curves connected to accelerated TRL increase
To assess overall potential impact of the process, incl. societal acceptance, techno-economic and ecological aspects with specific KPIs.
To evaluate potential social acceptance and to identify potential socio-technical and legal barriers for technology implementation in Europe.

Description

Task 7.1 Definition of selected reference cases and potential up-scale implementation scenarios (SOLVAY; EI, IRIS, TRSA) M6–M36

All partners jointly define the reference plant layouts and sizes according to WP4, WP5 and WP6 for technical potential process design upscale by elaboration of suitable plant components' capacity, interlinking dependencies, nominal mass and energy flows based on extensive process simulation (optimal process parameters) as well as main/neuralgic energy & materials needed for the plant components (electrochemical cell, operating materials, auxiliary materials and components, see also Task 4.1.3, 4.4, 5.2 and other related tasks). These data serve as input for an overall plant inventory, which is the basis for an integrated TEA and LCA. In addition, commercial up-scale implementation scenario(s) for representative EU countries (Germany, Spain) and markets (high volume/low concentration vice versa) according to typical electrochemical plant features and hydrogen peroxide market boundaries will be defined.

Task 7.2 Techno-economic assessment (TEA), evaluation of potential production sites, costs and inducible business cases. (EI; TRSA, IRIS, SOLVAY) M18–M46

Application scenarios (business cases) that are economically assessed based on task 7.1, also T6.3 will deliver valuable input to partial load operation. GIS-based analyses will be used to match suitable locations based on available renewable energy sources, water and infrastructure. To benchmark with competing renewable and fossil-based hydrogen peroxide technologies, a comprehensive TEA on product generation costs will be performed based on the scale-up defined in Task 7.1. Based on this spatial analysis, relevant market potentials of the technology will be estimated and used to project future CAPEX development in relation to learning and scaling effects (CoLLeCT model).

Task 7.3: Prospective time-resolved Life Cycle Assessment (LCA) (EI; TRSA, SOLVAY) M18–M47

The prospective time-resolved LCA will consider the production path of all components and the overall material composition, production processes, supply of inputs based on the energy and mass balance of defined reference cases in Task 7.1. Future application scenarios will be based on the project specific performance and material data (related to WP4, WP5 and WP6) and on representative databases like GaBi, GREET, Ecoinvent. EI will evaluate the GHG emissions/ neutrality and full environmental footprint of the entire value chain and define specific indexes such as electricity input/ mix to allow for the provision of KPIs related to the sustainability potential of the POWER2HYPE system, as well as comparisons with other technologies for the production of H2O2 via conventional anthraquinone oxidation (AO) process and alternative photochemical and thermal approaches.

Task 7.4: Stakeholder evaluation and societal and regulatory barriers analysis. (EI; all) M36–M48

To identify potential barriers for future implementations, EI, in cooperation with all partners, will assess relevant interest groups & stakeholders in terms of potential impacts by and acceptance of the POWER2HYPE technology roll-out. Sociological, technological, ecological, economic, and political dimensions of the technology are investigated using a multi-project proven Socio- and Multicriteria Model for reducing Barriers Analysis. Based on this socio-technic assessment supplemented by economic and ecologic implications from Task 7.2 and 7.3 and other tasks (e.g. Task 6.1 safety assessment), policy recommendations and measures for a smooth market entrance on European scale are elaborated to overcome barriers.

Work package WP8 – Dissemination, communication and exploitation

Work Package Number	WP8	**Lead Beneficiary**	11. LOBA
Work Package Name	Dissemination, communication and exploitation		
Start Month	1	**End Month**	48

Objectives

Maximizing the reach of POWER2HYPE results and findings
Ensuring wide visibility of the Process4Planet project POWER2HYPE
Providing the project with appropriate and effective communication media and tools
Supporting the POWER2HYPE consortium in the proactive management of IP generated by the project
Supporting the consortium in the exploitation of POWER2HYPE results, technologies and findings

Description

Task 8.1 Dissemination and communication activities (LOBA, all) M1-M48

Communication activities will have the role of increasing the visibility of the project in line with the communication strategy defined in the dissemination and communication plan. These activities will include among others the following (see section 2.2 for more details):

1. Dissemination and Communication plan: LOBA will develop the first version of the dissemination, communicationand exploitation plan at M6 and an updated version at M24.
2. Website: the website will be launched at M5 and maintained up to 5 years after the end of the project. A splash pagewill be available before the website is launched. The splash page will already contain the design of the website.
3. Promotional materials: LOBA will provide brochures, slides, posters, banners and GIFS for social media andpromotional videos.
4. Communication materials: LOBA will provide press releases, newsletters, design of training materials, ActionableKnowledge stemming from all WPs (such as factsheets, infographics, etc.)
5. Brand identity: the partners will create a unique brand universe to develop a community spirit, including brandguidelines, colours and font codes, and the logo variants

6. Participation in events, fairs, conferences and workshops: An overview of potential external events of interest for theproject is provided in section 2.2.
7. Organisation of pilot open days at Solvay's premises in Brussels: stakeholders will be invited to visit the POWER2HYPE pilot plan.
8. Organisation of project symposia (M24 & M48)- these events will be supported by a communication campaign andaimed at all target audiences. They will demonstrate a brief summary of all the work implemented during the project as well as the results achieved
9. Social Media: channels considered are LinkedIn, Facebook, Twitter and YoutubeThis task will also include the definition of the data management plan.

Task 8.2: IPR management and strategy (FHG, all) M1-M48

All the background brought by the partners has been identified at the proposal, although it will be thoroughly updated and specified as an integer part of the Consortium Agreement (CA), which will be signed before the Grant Agreement (GA) and thus before the starting date of the project. Of paramount importance for the CA is the definition of ownership of the project results (foreground) and IPR flows of each company involved in the project.
Using this as a starting point, the IP manager (FHG) will create a database to introduce the background and update IP during the project's period. The IP manager will be in charge of tracking and proposing adequate IPR actions for further technology and promising knowledge that could be generated during the project. FHG will also establish exploitation and dissemination strategies in line with the GA. The type of protection for the results (e.g., patent vs copyrights) will be analysed per outcome during the project and will condition the exploitation project outcomes. Every exploitable development will need the approval of the IP manager before it can be disseminated out of the reach of the consortium and the EC. Scientific/technical results articles/reviews are in general favoured and will be published in open access peer reviewed journals. If a potential patent may arise, FTO analysis will determine whether the research can be further pursued and exploited or whether there are IP barriers to overcome.

Task 8.3: Exploitation of project results (M4-M48) (Leader: LOBA, participants all):

This task will run in parallel with all project's activities, providing advice, guidance, fine-tuning and improvements. The Exploitation Manager (LOBA) will coordinate the following activities with the contribution of the partners: (i) Outline the main exploitable assets of the project to be produced as well as their availability time and the relevant target beneficiaries – WHAT; (ii) Identify the specific target groups that could benefit from project's outcomes – WHO; (iii) Provide an outline of the exploitation routes of the project assets anticipated for each one of them during the project and beyond – HOW/WHAT FOR. The exploitation Strategy will be delivered in D9.1 (M6), D9.2 (M24), D9.3 (M48) and will feed all the WPs. This task is highly interconnected with WP7, where concrete business cases will be explored. The exploitation plan will contain business plans for the key project outcomes.

LIST OF DELIVERABLES

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

The labels used mean:

Public — fully open (automatically posted online)

Sensitive — limited under the conditions of the Grant Agreement

EU classified —RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.1	Manual of management	WP1	1 - FHG	R — Document, report	SEN - Sensitive	3
D1.2	Set-up and operation of a secured project data management folder	WP1	1 - FHG	R — Document, report	SEN - Sensitive	4
D1.3	POWER2HYPE risk register	WP1	1 - FHG	R — Document, report	SEN - Sensitive	6
D1.4	Annual Reports including risk analysis	WP1	1 - FHG	R — Document, report	SEN - Sensitive	12
D2.1	Report on gas diffusion layer characterization and stability	WP2	9 - SGL	R — Document, report	SEN - Sensitive	9
D2.2	Membrane availability for demonstrator on the aromatic bipolar membranes proposed solution	WP2	1 - RHDO	R — Document, report	SEN - Sensitive	12
D2.3	Development of economically viable BDD electrodes for the electrolytic H2O2 production	WP2	4 - FAU	R — Document, report	SEN - Sensitive	18

D2.4	Catalyst formulation and the role of polarity and porosity on carbon	WP2	2 - WU	R — Document, report	SEN - Sensitive	22
D2.5	Report on the development of gas diffusion electrodes for H2O2 production	WP2	9 - SGL	R — Document, report	SEN - Sensitive	24

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D3.1	Report on Highly conductive, chemically and thermally stable AEMs and CEMs	WP3	2 - SSPI	R — Document, report	SEN - Sensitive	18
D3.2	Report on cathodic half-cell development and continuous cathodic H2O2 production	WP3	3 - WR	R — Document, report	SEN - Sensitive	24
D3.3	Continuous production of H2O2 by convergent paired electrolysis	WP3	1 - FHG	R — Document, report	SEN - Sensitive	30
D4.1	Report on the design for pilot electrolyser	WP4	5 - DIA	R — Document, report	SEN - Sensitive	24
D4.2	Definition of the layout for integrated electrolysis process available	WP4	1 - FHG	R — Document, report	SEN - Sensitive	28
D4.3	Report on upscaling & integration of concentrator	WP4	10 - SLVG	R — Document, report	SEN - Sensitive	30

D5.1	Report on basic plant design including: i) developed process design; ii) equipment specification; iii) preliminary operating manual	WP5	8 - TRSA	R — Document, report	SEN - Sensitive	28
D5.2	Delivery of the final pilot electrolyser ready to be integrated	WP5	5 - DIA	R — Document, report	SEN - Sensitive	33
D5.3	Pilot plant commissioning, manual of operation and testing campaign report	WP5	8 - TRSA	R — Document, report	SEN - Sensitive	36

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D5.4	Process monitoring and control with a digital twin	WP5	12 - IRIS	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	36
D6.1	HAZOP analysis report	WP6	7 - SOLVAY	R — Document, report	SEN - Sensitive	38
D6.2	Report on 1000h continuous operation	WP6	7 - SOLVAY	R — Document, report	SEN - Sensitive	44
D6.3	Report on operation flexibility	WP6	7 - SOLVAY	R — Document, report	SEN - Sensitive	48
D7.1	Definition and specification of evaluation scenarios of commercial deployment	WP7	7 - SOLVAY	R — Document, report	SEN - Sensitive	36

D7.2	Report on techno-economic assessment and site analysis	WP7	6 - EI	R — Document, report	SEN - Sensitive	46
D7.3	Report on GHG emissions and complete Life Cycle Assessment KPIs	WP7	6 - EI	R — Document, report	PU - Public	47
D7.4	Policy brief on socio-technical and regulatory barriers and recommended adjustments	WP7	6 - EI	R — Document, report	PU - Public	48
D8.1	Dissemination & communication plan activities and updates	WP8	11 - LOBA	R — Document, report	PU - Public	6
D8.2	Data Management plan	WP8	11 - LOBA	DMP — Data Management Plan	PU - Public	6
D8.3	IPR and Innovation Data base	WP8	1 - FHG	R — Document, report	SEN - Sensitive	12

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D8.4	Exploitation plan report and updates (incl. business plans)	WP8	11 - LOBA	R — Document, report	SEN - Sensitive	24

Deliverable D1.1 – Manual of management

Deliverable Number	D1.1	**Lead Beneficiary**	1. FHG
Deliverable Name	Manual of management		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	3	**Work Package No**	WP1
Description			
The manual of management is to act as a detailed guide providing directions on procedures and organization for implementation during the project.			

Deliverable D1.2 – Set-up and operation of a secured project data management folder

Deliverable Number	D1.2	**Lead Beneficiary**	1. FHG
Deliverable Name	Set-up and operation of a secured project data management folder		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	4	**Work Package No**	WP1
Description			
The setup and operation of a secured project data management folder is is key deliverable to be achieved. this will be carried out per the requirements of the relevant funding guidelines and to the satisfaction of the beneficiaries involved.			

Deliverable D1.3 – POWER2HYPE risk register

Deliverable Number	D1.3	**Lead Beneficiary**	1. FHG
Deliverable Name	POWER2HYPE risk register		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP1
Description			
The risk register will be established in the initial period in order to support tracking of the identified risks to the project.			

Deliverable D1.4 – Annual Reports including risk analysis

Deliverable Number	D1.4	**Lead Beneficiary**	1. FHG
Deliverable Name	Annual Reports including risk analysis		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	12	**Work Package No**	WP1
Description			
these reports are meant to contain the key and detailed information regarding the major progress of the project. Due date M12, M24 & M36 including a final report with risk analysis due M48			

Deliverable D2.1 – Report on gas diffusion layer characterization and stability

Deliverable Number	D2.1	**Lead Beneficiary**	9. SGL
Deliverable Name	Report on gas diffusion layer characterization and stability		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	9	**Work Package No**	WP2
Description			
the report will be prepared and delivered on time detailing the challenges and results of the diffusion layer's stability and characterization. specifically the impacts of implementing carbon/carbide based catalyst materials			

Deliverable D2.2 – Membrane availability for demonstrator on the aromatic bipolar membranes proposed solution

Deliverable Number	D2.2	**Lead Beneficiary**	1. RHDO
Deliverable Name	Membrane availability for demonstrator on the aromatic bipolar membranes proposed solution		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	12	**Work Package No**	WP2
Description			
This is a key deliverable to ensure the availability of the membranes that will be tested on schedule. The membranes is are essential for enabling ionic exchange, through various different concepts (aromatic proton, aromatic anion and bipolar membrane).			

Deliverable D2.3 – Development of economically viable BDD electrodes for the electrolytic H2O2 production

Deliverable Number	D2.3	**Lead Beneficiary**	4. FAU
Deliverable Name	Development of economically viable BDD electrodes for the electrolytic H2O2 production		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP2
Description			
Making the BDD electrodes for the electrolytic H2O2 production economically viable is critical step in achieving the overall goals of the project and as such is set as a deliverable to be achieved in the by the given deadline in order to allow for sufficient time to complete the remaining steps.			

Deliverable D2.4 – Catalyst formulation and the role of polarity and porosity on carbon

Deliverable Number	D2.4	**Lead Beneficiary**	2. WU
Deliverable Name	Catalyst formulation and the role of polarity and porosity on carbon		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	22	**Work Package No**	WP2

Description
Refining the Catalyst formulation and establishing (and quantifying) the relationship of the of polarity and porosity on carbon is necessary for performance goals

Deliverable D2.5 – Report on the development of gas diffusion electrodes for H2O2 production

Deliverable Number	D2.5	**Lead Beneficiary**	9. SGL
Deliverable Name	Report on the development of gas diffusion electrodes for H2O2 production		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP2
Description			
A subsequent report on the on the development of gas diffusion electrodes for H2O2 production will be prepared. this will build up the report from D2.1 as the accomplishments of that deliverable will make achieving this one possible.			

Deliverable D3.1 – Report on Highly conductive, chemically and thermally stable AEMs and CEMs

Deliverable Number	D3.1	**Lead Beneficiary**	2. SSPI
Deliverable Name	Report on Highly conductive, chemically and thermally stable AEMs and CEMs		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP3
Description			
This report will detail the functional and material properties of the various membranes targeted for development			

Deliverable D3.2 – Report on cathodic half-cell development and continuous cathodic H2O2 production

Deliverable Number	D3.2	**Lead Beneficiary**	3. WR
Deliverable Name	Report on cathodic half-cell development and continuous cathodic H2O2 production		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP3
Description			
This report will be on optimization of the needed parameters for the anodic water oxidation reaction (WOR). It will also detail progress in optimizing reaction conditions, the nature of the anode and design of the anodic half-cell will be essential to determine the stability and performance indicators.			

Deliverable D3.3 – Continuous production of H2O2 by convergent paired electrolysis

Deliverable Number	D3.3	**Lead Beneficiary**	1. FHG
Deliverable Name	Continuous production of H2O2 by convergent paired electrolysis		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive

Due Date (month)	30	**Work Package No**	WP3
Description			
This document will detail the continuous production of H2O2 by convergent paired electrolysis achieved at this point. This being the ultimate end-goal of WP3 as it marks a major step in the completion of a working system			

Deliverable D4.1 – Report on the design for pilot electrolyser

Deliverable Number	D4.1	**Lead Beneficiary**	5. DIA
Deliverable Name	Report on the design for pilot electrolyser		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP4
Description			
This report on the design for pilot electrolyser will be meant to detail its performance in a low-resistance, high-efficiency setup. it will summarize achievements needed for the electrolyser to be implemented in the demonstrator and the prototype.			

Deliverable D4.2 – Definition of the layout for integrated electrolysis process available

Deliverable Number	D4.2	**Lead Beneficiary**	1. FHG
Deliverable Name	Definition of the layout for integrated electrolysis process available		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	28	**Work Package No**	WP4
Description			
This document will contain the design of the prototype electrolyser as well as the layout for a suitable gas and fluid flow and operation parameters for the paired electrolysis.			

Deliverable D4.3 – Report on upscaling & integration of concentrator

Deliverable Number	D4.3	**Lead Beneficiary**	10. SLVG
Deliverable Name	Report on upscaling & integration of concentrator		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	30	**Work Package No**	WP4
Description			
This report will essentially contain the layout and guidelines for the process for the combined H2O2 production and a mass flow diagram that will be the basis for the pilot system in WP5.			

Deliverable D5.1 – Report on basic plant design including: i) developed process design; ii) equipment specification; iii) preliminary operating manual

Deliverable Number	D5.1	**Lead Beneficiary**	8. TRSA

Deliverable Name	Report on basic plant design including: i) developed process design; ii) equipment specification; iii) preliminary operating manual		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	28	**Work Package No**	WP5
Description			
This document will contain details on developed process design and equipment specification as well as preliminary operating manual. These are essential elements in order to deliver a finalized pilot design			

Deliverable D5.2 – Delivery of the final pilot electrolyser ready to be integrated

Deliverable Number	D5.2	**Lead Beneficiary**	5. DIA
Deliverable Name	Delivery of the final pilot electrolyser ready to be integrated		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	33	**Work Package No**	WP5
Description			
A document with design and performance details of the pilot electrolyser ready to be integrated			

Deliverable D5.3 – Pilot plant commissioning, manual of operation and testing campaign report

Deliverable Number	D5.3	**Lead Beneficiary**	8. TRSA
Deliverable Name	Pilot plant commissioning, manual of operation and testing campaign report		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP5
Description			
The deliverable will be a detailed report on the pilot plant commissioning which will include the manual of operation needed for implementation as well as testing campaign report			

Deliverable D5.4 – Process monitoring and control with a digital twin

Deliverable Number	D5.4	**Lead Beneficiary**	12. IRIS
Deliverable Name	Process monitoring and control with a digital twin		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP5
Description			
The deliverable shall demonstrate functioning process, monitoring and control with a digital twin. The achievement of this deliverable will mean the commissioning of the Demo unit at Solvay site. Furthermore as part of the functioning process (as well as monitoring and control), operation and safety trainings will be given to the operators.			

Deliverable D6.1 – HAZOP analysis report

Deliverable Number	D6.1	**Lead Beneficiary**	7. SOLVAY
Deliverable Name	HAZOP analysis report		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	38	**Work Package No**	WP6
Description			
This report will contain a HAZOP methodology assessment. It will also provide insights on the components of the demonstration unit.			

Deliverable D6.2 – Report on 1000h continuous operation

Deliverable Number	D6.2	**Lead Beneficiary**	7. SOLVAY
Deliverable Name	Report on 1000h continuous operation		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	44	**Work Package No**	WP6
Description			
Report this report will provide details on operational challenges, performance, learnings specifically about the the electricity and chemicals consumption, peroxide concentration and quality, stability tests			

Deliverable D6.3 – Report on operation flexibility

Deliverable Number	D6.3	**Lead Beneficiary**	7. SOLVAY
Deliverable Name	Report on operation flexibility		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP6
Description			
This report will aim to elaborate on the evaluation done of the operational range of the unit. Specifically, this will be done in terms establishing various operational ranges, meaning the minimum and maximum hydrogen peroxide concentration achievable, minimum and maximum current density achievable, minimum and maximum operational temperature and the minimum and maximum flow rates achievable.			

Deliverable D7.1 – Definition and specification of evaluation scenarios of commercial deployment

Deliverable Number	D7.1	**Lead Beneficiary**	7. SOLVAY
Deliverable Name	Definition and specification of evaluation scenarios of commercial deployment		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP7
Description			

This document will define the reference plant layouts and sizes for technical potential process design upscale by elaboration of suitable plant components' capacity, interlinking dependencies, nominal mass and energy flows based on extensive process simulation as well as main/neuralgic energy & materials needed for the plant components

Deliverable D7.2 – Report on techno-economic assessment and site analysis

Deliverable Number	D7.2	**Lead Beneficiary**	6. EI
Deliverable Name	Report on techno-economic assessment and site analysis		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	46	**Work Package No**	WP7
Description			
The document that constitutes this deliverable will detail the inputs determined for partial load operation. It will show the methodology used to match suitable locations based on available renewable energy sources, water and infrastructure.			

Deliverable D7.3 – Report on GHG emissions and complete Life Cycle Assessment KPIs

Deliverable Number	D7.3	**Lead Beneficiary**	6. EI
Deliverable Name	Report on GHG emissions and complete Life Cycle Assessment KPIs		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	47	**Work Package No**	WP7
Description			
This document will provide the information gathered in order to evaluate the GHG emissions/neutrality and full environmental footprint of the entire value chain as well as define specific indexes such as electricity input/mix to allow for the provision of KPIs related to the sustainability potential of the POWER2HYPE system. It will also contain information on comparisons with other technologies for the production of H2O2 via conventional AO process and alternative photochemical and thermal approaches.			

Deliverable D7.4 – Policy brief on socio-technical and regulatory barriers and recommended adjustments

Deliverable Number	D7.4	**Lead Beneficiary**	6. EI
Deliverable Name	Policy brief on socio-technical and regulatory barriers and recommended adjustments		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP7
Description			
This document will expound upon the barriers identified for future implementations, In addition it will provide an assessment of relevant interest groups & stakeholders in terms of potential impacts by and acceptance of the POWER2HYPE technology roll-out. The scope will include sociological, technological, ecological, economic, and political dimensions of the technology are investigated using a multi-project proven Socio- and Multicriteria Model for reducing Barriers Analysis.			

Deliverable D8.1 – Dissemination & communication plan activities and updates

Deliverable Number	D8.1	**Lead Beneficiary**	11. LOBA

Deliverable Name	Dissemination & communication plan activities and updates		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP8
Description			
The point of this deliverable is the visibility of the project in line with the strategy defined in the dissemination and communication plan. M6, M12, M24, M36			

Deliverable D8.2 – Data Management plan

Deliverable Number	D8.2	**Lead Beneficiary**	11. LOBA
Deliverable Name	Data Management plan		
Type	DMP — Data Management Plan	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP8
Description			
the data management plan is a key document for a project. it will determine the internal and external (to the consortium) practices for communication and storage of data generated. This will among other things affect the IPR and innovation database. M6, M24, M48			

Deliverable D8.3 – IPR and Innovation Data base

Deliverable Number	D8.3	**Lead Beneficiary**	1. FHG
Deliverable Name	IPR and Innovation Data base		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	12	**Work Package No**	WP8
Description			
Every exploitable development will (after the approval of the IP manager) be included in the database and disseminated out of the reach of the consortium and the EC.			
Exploitable developments include but are not limited to patents, industrial designs, works of authorship and so on. The database is meant to track and store key relevant information about each item acting as a key document for internal reference. M12, M24, M36, M48			

Deliverable D8.4 – Exploitation plan report and updates (incl. business plans)

Deliverable Number	D8.4	**Lead Beneficiary**	11. LOBA
Deliverable Name	Exploitation plan report and updates (incl. business plans)		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive

Due Date (month)	24	**Work Package No**	WP8
Description			
Exploitation plan report and updates are key documents that will be prepared under the Exploitation Manager (LOBA) who will coordinate the outline the main exploitable assets of the project to be produced as well as their availability time and the relevant target beneficiaries			

LIST OF MILESTONES

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
1	Catalyst and EC Process	WP2, WP3	9-SGL	Catalysts enable a continuous electrochemical process for at least 100h with FE>80% and J>400 mAcm2	24
2	Definition of the catalysts formulation	WP2	9-SGL	Review of deliverable D2.3 and 2.4	24
3	Electrochemical process validation at lab scale	WP4, WP2, WP3	3-WR	Demonstration of a process in lab scale with efficient catalysts (160 % total FE). Review of deliverable 3.3.	24
4	Integration & testing of H2O2 concentrator at SolvGE facility completed	WP4	5-DIA	Testing and internal demonstrations with varying operational conditions successfully conducted data are available. Review of deliverable D4.3	29
5	Reference plant layouts and scales defined	WP7	6-EI	Reference plant layouts available	30
6	Plant is commissioned, conditioned for shipping and ready to be shipped to Solvay's site	WP5	8-TRSA	Preliminary functionality tests of the running plant are proved. Review of deliverable D5.3	36
7	Learning curve and scaling effects determined for the core technologies	WP7	6-EI	For all core technologies involved in POWER2HYPE key scaling effects for the further commercialization steps would have been assessed and addressed	40
8	Pilot and Stacks	WP5, WP4	5-DIA	The stacks are manufactured and the pilot plant is operating with targets of Milestone 1.	36
9	End of the plant testing over continuous run for 1000 h	WP6	7-SOLVAY	Review deliverable D6.2	44

10	Final validation of the Pilot plant results	WP6	7-SOLVAY	Review deliverable D6.3	48

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
	including validation of the H2O2 produced with end-users				

LIST OF CRITICAL RISKS

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
1	Substantial delay or lack of performance by a partner (low likelihood, medium severity)	WP1	Close monitoring of the progress periodically (every 6 months). Potential reallocation of resources if deemed necessary.
2	Partner leaves consortium (low likelihood, high severity)	WP1	General Assembly will meet and decide on appropriate measures to ensure that the project's objectives can be met; preferred option is to re-allocate work to another capable partner (see Capacity of participants) in consortium; alternative options are subcontracting (if scope is suitable for sub-contracting) or a competitive call for a new partner.
3	Gas diffusion layer (MPL sublayer) stability against hydrogen peroxide insufficient (low likelihood, medium severity)	WP2	MPLs based on graphitic carbons and/or conducting oxides
4	Aromatic backbone ionomers family not sufficiently chemically stable (low likelihood, medium severity)	WP2, WP3	Pivot our efforts on per fluorinated backbones under development in WP3.

5	AEM aromatic ionomer shows only insufficient anion conductivity (medium likelihood, medium severity)	WP2, WP3	Increase the charge carrier (decrease EW) in the ionomer backbone to improve conductivity

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
6	A mismatch in e.g. the ion transport with the integration of the anodic and cathodic process (medium probability, medium severity)	WP3	Consider changing the pH of one half-cell, utilize a bipolar membrane or use a mediator, which works under different conditions.
7	Efficient decomposition of H2O2 from exhaust to H2O before letting out to the atmosphere (medium likelihood, medium severity).	WP5, WP4	Effective catalytic decomposition will be used to mitigate the risk of letting the exhaust out with trace of H2O2 to the atmosphere.
8	Misfunctioning of electrochemical stack in pilot plant (Low likelihood, high severity)	WP5	A thorough set of tests will be realized during WP4 prior to pilot start to assess performance of the stack and ensure optimal functioning for pilot and demo.
9	Stack design does not produce electrolyte free H2O2 stream (low likelihood, high severity)	WP5, WP4	A unit for electrolyte separation from the H2O2 stream will be considered in the design of the plant.
10	Stack gets broken during the tests (Medium likelihood, high severity)	WP5	Two stacks will be constructed by DiaCCon in case one results broken during piloting or demo stages.
11	Reliability of the results from WP 7 strongly depends on the extent of process data availability and whether data quality requirements are met. The precision, completeness and representativeness of the data are relevant to generate reproducibility. (Medium likelihood, high severity)	WP6, WP5, WP7, WP4	A detailed questionnaire for data transfer from other WPs is foreseen and filling data gaps is a steadily process throughout the project. Missing data will be replaced with data from previous work packages (earlier stage within the project) or replaced with (assumptions based on) literature data.

12	Uncertainty of the information and data. (Medium likelihood, medium severity)	WP6, WP5, WP7, WP4	The modelling with robust software (e.g., GaBi) guarantees fast updates and generates reproducibility, also, sensitivity analyses will be carried out
13	Few information available in existing literature, e.g. in regard to socio-technical aspects (High likelihood, medium severity)	WP7	Literature references of and comparisons with comparable products and technologies will be made, remaining knowledge gaps will be used to refine further R&D activities and recommendations (see also D7.4)

PROJECT REVIEWS

Project Reviews

Grant Preparation (Reviews screen) — Enter the info.

Review No	**Timing** (month)	**Location**	**Comments**
RV1	18		
RV2	36		
RV3	48		

TABLE OF HISTORY OF CHANGES	
Version (date)	Changes
First GAP submission	**Annex 1 Part A**
	-WP1 description: change to the deadline for signing the CA to "**before the signature of the grant agreement**" - Project management (WP1) related risks added to the list of critical risks.
	Annex 1 Part B
	- State-of-the-art on ORR and cell design updated in Section 1.1.2, page 6, 7 -State of the art on WOR updated in Section 1.1.2, page 6, 7. - Added (semi)quantitative metrics with respect to contribution to Sustainable Development Goals, Section 2.1.1, page 19, 20. -Added Ethics section 4 , Section 4.1, Page 41. - Added text on the importance of load flexibility in a future 100% renewable energy landscape with inherent intermittency of power supply, Section 1.2.1, page 11. - **Purchase costs table** -FGH: changed text to test -FGH: detail added for the COMSOL, Costs for Comsol (FuelCell+Electrolyser (8,995) and Heat Transfer module (6,995)). This is a one-time fee for permanent license without support. -IRIS: added the clarification for othrt goods, works and services. Also the double appearance of 40keuro removed. Section 3.1, Page 36 -TRSA: added clarification for shipping -WU, LOBA, IRIS, TRSA clarifications/additional details provided as needed -SLVG clarifications/additional details provided as needed **-** **Subcontracting costs table:** -SGL discrepancy with the sub-contracting costs vs Annex 2 budget, was corrected
Second GAP submission	**Annex 1 Part A**
	- Anthraquinone oxidation (AO) process has been defined in Excellence as well as in WP7 description, Task 7.3 - WP2 description: The PM for partner FAU has been corrected from 36 to 42 - The description of deliverables completed for deliveables: D1.4, D2.2, D8.1, D8.2, D8.3 and D8.4 - The dissemination level of deliverables D7.1, D7.2, D8.1 and D8.4, changed to 'SEN – Sensitive. -WP3 description: study of load flexibility added to T3.4 - SGL Carbon replaced with SGL Fuel Cell Components GmbH
	Annex 1 Part B

	- Elaborated the barriers for implementation in Sec 2.1.3, page 22 - Gantt chart has been updated, Section 3.1, page 33 -- SGL Carbon replaced with SGL Fuel Cell Components GmbH in Section 1.1.2, Page 7 and Section 3.2, Page 39 - **Purchase costs table:** - the beneficiary numbering has been updated
	- WU updates integrated (The additional budget amounts Associated with docu Ares(2022)7691164 - 08/11/2022 were elaborated upon as requested) Sectior Page 34. -FAU : added clarification for equipment and other goods, works and services. Section 3.1, Page 35. -DIA : added clarification and elaborated the costs in other goods, works and services part (Section 3.1, Pge 35).
Third GAP submission	**Annex 1 Part A**
	-The description of deliverables completed for deliverables: D6.3, and D8.3
	Annex 1 Part B
	-Figure 15 changed (MSA and MSB, renamed to MS1 and MS8), Section 3.1, page 32 -Gantt chart has been updated Section 3.1, page 33 -Section 1.2.2, page 14, Milestone 2 changed to milestone 8 (earlier milestone (MSB) was wrongly mentioned as milestone 2). With the new update (according to the Part A, the milestone MSB was MS8. **- Purchase costs table** -TRSA beneficiary number in Purchase costs table corrected Section 3.1, Page 36 -WU further breakdown provided in purchase costs table Section 3.1, page 34 -SGL: purchase costs tables and internally invoiced tables updated Section 3.1, page 37 -TRSA : updated description regarding insurances, Section 3.1, page 36 - DIA: Purchase cost table: Description of Other goods, works and services cost items for partner DIA revised (cost items corrected to sum up to EUR 65,000) Section 3.1, page 35. -SLVG: updated purchase costs with clearer and more detailed descriptions **- Ethic Section** - Ethics Issue table updated ((Personal data, Environment, Health and Safety and AI) to Section 4.1, page 41.

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1. Excellence

In line with the European Green Deal, the Processes4Planet partnership and the European initiative for more affordable, secure and sustainable energy (REPowerEU), the RIA proposal POWER2HYPE aims to establish a **sustainable route for the hydrogen peroxide production.** Based on air and water as feedstocks, and renewable energy as the sole energy source, the POWER2HYPE route addresses an economically viable and green alternative to the classical energy-demanding anthraquinone oxidation (AO) process. Hydrogen peroxide will be produced by paired electrolysis at both the cathode and anode using an innovative custom-made 200% electrolyzer. Aiming the **electrification and decentralization of the chemical industry**, the POWER2HYPE concept proposes the **TRL 6 demonstration of electrolytic hydrogen peroxide production at any given concentration (from 20 to 99 %) at the site where it is needed**. With this flexibility of concentration range, hydrogen peroxide could be used from chemical intermediates, sanitation, bleaching to niche markets as propulsion fuels.

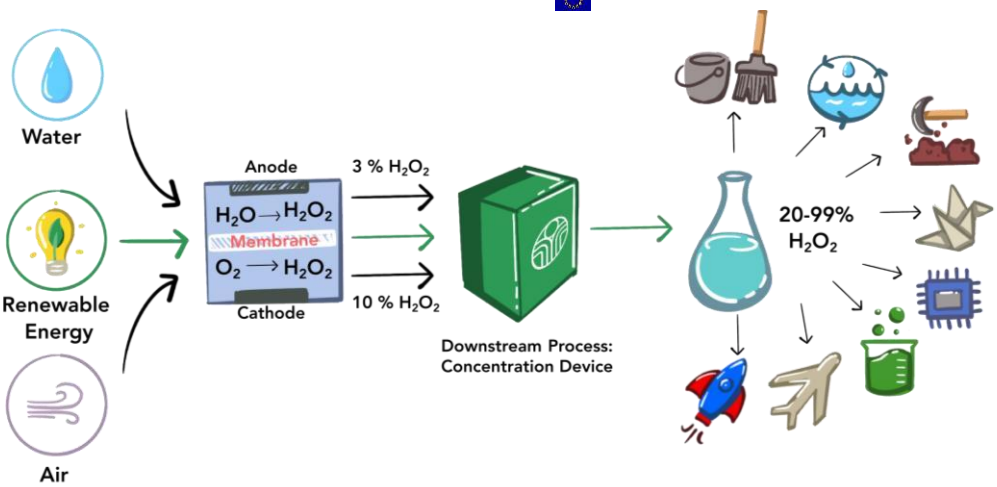


*Figure 1: **POWER2HYPE** concept for electrochemical H_2O_2 production and downstream process with a final concentration of 20 to 99% H_2O_2.*

Hydrogen peroxide (H_2O_2) is a widely used commodity chemical with a market evaluated at 3.6 billion € in 2020 and forecasted to grow to 4.7 billion € by 2026.[1] It has a strong oxidizing capacity, and releases only H_2O and O_2 as sideproducts. Industrial applications include wastewater treatment,[2] bleaching,[3] sanitation,[4] chemical synthesis[5] and aerospace fuels.[6] Thus, **H_2O_2 is listed among the 100 most essential chemicals globally**.[6] The **H_2O_2 production capacity in Europe reached 1.2 million tons** (calculated as 100 % H_2O_2) in 2020, i.e. 35 % of the global production.[7] Worldwide H_2O_2 production is expected to continue its growth trend, reaching 7 Mtons/y by 2030.[8] POWER2HYPE **partner Solvay, as leading global H_2O_2 producer and supplier, commands half of the current whole European production capacity alone.[9]**

The current industrial production of H_2O_2 is predominantly (95%) based on the **alkylanthraquinone autoxidation process (AO)**,[10] which is **considerably energy demanding** (4.4 kWh/kg H_2O_2,[11] mostly coming from fossil fuels and amounting up to 6.2 10^9 kWh/year of energy consumption), leading to significant GHG emissions represent **2.4 million tons CO_2,** [13] (i.e. 1.8%, to the total GHG of the chemical industry in Europe)[12]. Additionally, this process is highly **centralized in large chemical units**, limiting the supply of H_2O_2 to remote areas. Finally, the process **involves sequential hydrogenation and oxidation steps of alkylanthraquinone (AQ)**, which is dissolved in a mixture of organic solvents, followed by H_2O_2 extraction and distillation, with the **subsequent generation of waste**. Compared to the classical AO process, the new electrochemical route proposed by POWER2HYPE will: (i) achieve better overall **energy efficiency** (energy demand of 3 kWh/kg of H_2O_2), (ii) use **sustainable and abundant materials** as catalysts and cell components; (iii) leave **no subsequent generation of toxic waste**/side products (vs. highly hazardous products in AO); (iv) use a low-energy downstream process (DSP) to produce hydrogen peroxide at **any given concentration** up to 99 wt.%; (v) improve the dynamics/agility of the process and large volume down-stream products (e.g. propylene oxide) in order to adapt to the intermittent availability of **renewable electricity** quickly; (iv) **decentralize** the hydrogen peroxide production and even in **miniaturized** and therefore **cost-effective and lowerbudget** production lines, taking in account the flexibility of electrochemical devices and its integration with widespread distribution of renewable electricity.

[1] Global Hydrogen Peroxide Market Outlook. https://www.expertmarketresearch.com/reports/hydrogen-peroxide-market (accessed 19.03.2022).
[2] Horová, D. et al., Synthesis of MFI structured iron silicates and their catalytic performance in phenol removal by wet peroxide oxidation. *React. Kinet. Mech. Catal.* **2020,** *130* (2), 1077-1092.
[3] Hage, R. et al., Applications of transition-metal catalysts to textile and wood-pulp bleaching. *Angew. Chem. Int. Ed. Engl.* **2005,** *45* (2), 206-22.
[4] Rubio-Romero, J. C. et al., Disposable masks: Disinfection and sterilization for reuse, and non-certified manufacturing, in the face of shortages during the COVID-19 pandemic. *Saf. Sci.* **2020,** *129*, 104830.
[5] Russo, V. et al., Chemical and Technical Aspects of Propene Oxide Production via Hydrogen Peroxide (HPPO Process). *Ind. Eng. Chem. Res.* **2013,** *52* (3), 1168-1178. [6] Gohardani, A. S, et al. Green space propulsion: Opportunities and prospects. *Prog. Aerosp. Sci.* **2014,** *71*, 128-149.
[6] Myers, Richard L. *The 100 most important chemical compounds: a reference guide*. ABC-CLIO, 2007.
[7] https://eippcb.jrc.ec.europa.eu/sites/default/files/2019-11/JRC109279_LVOC_Bref.pdf
[8] https://www.marketresearch.com/ChemAnalyst-v4204/Global-Hydrogen-Peroxide-Plant-Capacity-14810413/ (accessed on 19.03.22)
[9] https://www.solvay.com/en/chemical-categories/peroxygens/hydrogen-peroxide-h2o2
[10] Edwards, J. K. et al ., Palladium and gold-palladium catalysts for the direct synthesis of hydrogen peroxide. *Angew. Chem. Int. Ed. Engl.* **2008**, 47 (48), 9192-8.
[11] Hargreaves, J.S.J., et al. "Minimizing energy demand and environmental impact for sustainable NH3 and H2O2 production-A perspective on contributions from thermal, electro-, and photo-catalysis." Applied Catalysis A: General 594 (2020): 117419. [13] Based on 1.19 kg CO_2/kg H_2O_2 (EPA/600/R-16/176, August 2017)
[12] Bazzanella, A., & Ausfelder, F. (2017). *Low carbon energy and feedstock for the European chemical industry: Technology Study*. DECHEMA, Gesellschaft für Chemische Technik und Biotechnologie eV.

1.1 Objectives and ambition

The **POWER2HYPE** objective is to develop and demonstrate a new process for hydrogen peroxide production, changing the established, energy-demanding chemical route by a sustainable electrochemical one.

1.1.1 Objectives

The specific objectives of the project are directly extracted from the necessary improvements necessary for the process State-of-the Art, as can be seen in the following table.

Table 1: Anthraquinone and POWER2HYPE processes.

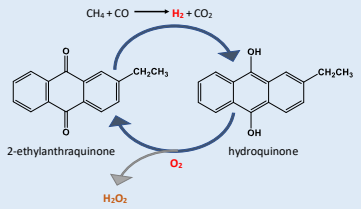
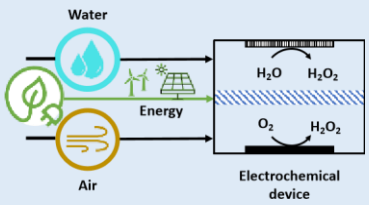
AO Process	**POWER2HYPE**
	
Energy demanding (3.8-4.6 kWh/kg H_2O_2)[8]	Energy efficient (3 kWh/kg H_2O_2 - up to 33% energy saving)
Noble-metal catalyst materials (palladium-based)	Abundant catalysts and components (carbon-based)
Predominantly centralized	Modular prototype allowing flexibility / decentralization
Hydrogenation of AQ with H_2 (from methane)	Requires no input of hydrogen (no gases for hydrogenation)
Use of organic solvents for AQ dissolution and extraction	Use of only water-based electrolytes
Thermal H_2O_2 distillation to 50 and 70 wt.%	Customized low-energy DSP reaching concentrations of up to 99 wt.% H_2O_2
Waste products (waste stream of quinone-derived byproducts) and emission of alkylated aromatics as volatile organic compounds (VOCs).	Possibly only H_2O and O_2 as non-toxic side- or wasteproducts, no use of solvents
Risks of overpressure, catalyst breakthrough from hydrogenation, and explosions during oxidation	No steps under pressure, no risk of explosions during the process, reaction can be stopped by switching of the electrical power supply
Emissions must be controlled associated with the oxidation step, drying of the working solution and the hydrogenation step	No intrinsic emissions in the process
High CO_2 emissions in the overall industrial process due to the use of fossil energy sources	Decrease the release of direct CO_2 emissions by 100% due to the use of renewable energy sources

Table 2: Objectives of POWER2HYPE with KPIs.

OB1: Development and optimization of cell components for large-scale electrochemical production of hydrogen peroxide
• **Carbon-based catalysts materials and gas-diffusion layers** for the cathodic Oxygen Reduction Reaction (ORR). • **Boron-doped diamond layers on low-cost substrates** for the anodic Water Oxidation Reaction (WOR). • The novel electrodes should enable a continuous H_2O_2 production with Faradaic Efficiency (FE) > 80% at each half-cell with current densities J > 400 mA cm^{-2} for at least 1000 h (prototype-scale). • chemical-resistant and highly conductive **ion exchange membranes** for the efficient coupling of ORR with WOR resulting in a low energy consumption.
OB2: Design, development and integrating a new electrochemical process and reactor

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The different steps of the process will be: • **An electrochemical cell reactor with tailor-made carbon-based components**, optimized fluid flow path and manifold scheme, and maximum energy consumption of 3 kWh/kg H_2O_2. • **A paired electrochemical process** with cathodic **and** anodic H_2O_2 generation with total **FE > 160 %** in smallscale 10 cm^2 lab cells as well as in scaled-up and **stacked 200 cm^2 pilot electrolysers**. • **A low-energy concentration device**, capable of increasing H_2O_2 concentration from 3-10% to any given concentration up to 99 wt.% H_2O_2.
OB3: Piloting and demonstrating the innovative electrochemical process in a relevant environment
To design, produce, install, and commission a **pilot scale continuous flow demonstrator** capable of producing up to **20 kg H_2O_2/day (based on 100% H_2O_2)** in a relevant industrial environment. The prototype will be placed in a container to provide it with mobility and for decentralized operation, and demonstrated in Brussels (Belgium), in Solvay's Research and Technology labs.
OB4: Evaluating the feasibility and sustainability of the new process, from the technical, environmental and (socio-)economic perspectives
• A **sophisticated Digital Twin (DT)** of the physical pilot will enable monitoring, analyzing and controlling the process by the generation of real-time data of the process parameters. **The data generated will pave the way to understand scale-up requirements.** • **Life-cycle assessment** (LCA) and **techno-economic analysis (TEA)** conducted to validate industrial feasibility and economic viability of the POWER2HYPE process.
OB5: Create and implement an exploitation strategy for the project developments
• The **commercialization of the new process as a whole and all the components** designed by POWER2HYPE technology developers. Business plans will be produced in order to develop the business cases presented in the proposal. • The **marketability of hydrogen peroxide produced in a sustainable way**. • **Further research and spin-off projects** coming from POWER2HYPE.
OB6: Ensure the dissemination of the project results and the promotion of the project, through ad-hoc strategies that target POWER2HYPE audience and key stakeholders.

1.1.2 Project ambition & added value vs. state of the art (SoA)

The anthraquinone oxidation (AO) process

The AO process is a multistep process that requires significant energy input and a well-developed infrastructure. Moreover, it poses substantial risks associated with oxidation steps under pressure and the handling of concentrated hazardous chemicals like H_2O_2.[13] This process can be divided into 4 steps, starting with the (i) hydrogenation of 2alkylanthraquinone (AQ) with H_2 to anthrahydroquinone in a mixture of solvents using Pd-based catalysts. The resulting anthrahydroquinone undergoes (ii) oxidation by O_2 to generate H_2O_2 and so is anthraquinone regenerated. This oxidation step is highly hazardous and requires a careful process design for risk mitigation.[14] Subsequently, the (iii) extraction with solvents removes H_2O_2 and the anthraquinone left in the organic solvent is recycled. This extraction step is crucial for the economic viability of the whole process, and unextracted peroxide can account for big losses in the AO process. Finally, (iv) crude H_2O_2 is purified and distilled to reach the usual commercial grade concentration of 30-70 wt.% H_2O_2.

Considering all the specific steps and the risks involved, besides considerable improvements and high process efficiency in large scales, the production of H_2O_2 through **this indirect AO process is rather energy demanding** with a consumption of 2.9 - 5.7 MJ/kg_{H2O2} (0.8 – 1.6 kWh/kg_{H2O2})[15] plus 3.0 kWh/kg_{H2O2} associated with the generation of H_2.[16] Moreover, highly concentrated H_2O_2 solutions **require special safety measures** for handling, transportation, and storage. In this regard, besides specific applications requiring high concentrated H_2O_2, the main use of H_2O_2 in bleaching and disinfectant requires 2-8 wt.% H_2O_2 solutions, with the consequence, that H_2O_2 is diluted again after shipped at 50-70%. Transportation in lower (safer) concentrations has the drawback of moving mostly water, thus negatively affecting the carbon footprint and economic viability of the final product. Therefore, the possibility of providing a direct and decentralized process for the H_2O_2 synthesis would be safer, cheaper, and significantly more sustainable not only for the chemical industry but also for the society.

[13] Fortunato, G. V. et al., The oxygen reduction reaction on palladium with low metal loadings: The effects of chlorides on the stability and activity towards hydrogen peroxide. *J. Catal.* **2020,** *389*, 400-408.
[14] Goor, G. et al., Hydrogen Peroxide. *In Ullmann's Encyclopedia of Industrial Chemistry*, **2022**, pp 1-40.
[15] Falcke et al., (2017) European Union – Best Availbale Techniques (BAT) Reference Document for the production of Large Volume Organic Chemicals, Page 562.
[16] Based on the stochiometric use of H_2 and an specific energy usage of 50 kWh/kg H_2 by electrolysis (IRENA (2020) Green Hydrogen Cost reduction – Scaling up Electrolysers)

Although some effort in the decentralization of the AO process has been showcased by Solvay,[17] the multistep chemical process has inherent drawbacks, as shown in Table 1.

As renewable energy is becoming increasingly affordable, the key challenge is to transform electrical energy into energy-dense chemical bonds. Approaches using photoelectrochemical processes[18] are valuable possible routes from fundamental research, however, considering the decarbonization goals of the European 2030 climate and energy framework, [19] it will be particularly challenging to reach industrialization on time due to their low TRLs, low conversion efficiencies (few % solar-to-chemical) and serious issues with collecting products that are generated over considerable surface areas.

Electrochemical production of hydrogen peroxide

The electrochemical production of H_2O_2 is based on water, air, and renewable electric energy as input. Thus, it is also representing an elegant green solution for onsite and on-demand H_2O_2 production, thereby avoiding risks associated with large-scale storage and transportation of highly concentrated H_2O_2. There are two electrochemical pathways for H_2O_2 production: the **cathodic oxygen reduction reaction (ORR, Eq. 1)** and the **anodic water oxidation reaction (WOR, Eq. 2)**. Both pathways involve a two-electron transfer:[20]

$$O_2 + 2H^+ + 2e^- \rightleftharpoons H_2O_2 \qquad E^0 = 0.67\ \text{V } vs.\ \text{RHE} \qquad (1)$$

$$2H_2O \rightleftharpoons H_2O_2 + 2H^+ + 2e^- \qquad E^0 = 1.76\ \text{V } vs.\ \text{RHE} \qquad (2)$$

ORR and WOR reactions have already been explored as half-cells. **The ambition of** POWER2HYPE **is to match both into one single electrochemical cell**, thereby making optimal use of the electrons and obtaining H_2O_2 as main product at the anode and cathode.

The **cathodic pathway (Eq. 1)** has already been known for almost 150 years[23] and reached a pre-commercial scale in the 1990s for the on-site production of alkaline H_2O_2, utilizing carbon chips coated with a high surface area carbon/PTFE composite.[24] Major disadvantage was the alkaline environment in which H_2O_2 decomposes more readily,[21] and limits its application due to intensive DSP to obtain pure H_2O_2. As for acidic electrolytes, noble metalbased (Pd-Hg, Au-Pd and Au-Pt-Ni) are the most active catalyst materials. [22,23] However, the scarcity of the platinumgroup metals and the environmental issues associated with Hg make the use of these type of catalysts questionable regarding sustainability. Carbon-based catalysts show relatively low activity under acidic conditions, however high activity and faradaic efficiency under alkaline conditions,[22,26,27] which matches the cathodic environment in a dual membrane configuration.[37] For the **improvement of the cathodic catalyst materials,** POWER2HYPE aims for catalysts based on carbon with no noble metals. In POWER2HYPE, the interplay between the nature of the carbon, its surface chemistry and porosity, effect of a dopant and non-noble metal species on the performance of the carbon in H_2O_2 production will be investigated.

The ORR involves a ternary interface reaction of gas (O_2), liquid (an aqueous electrolyte) and solid (active catalysts). Such ternary interface reactions require the use of **gas diffusion layers (GDL)**. The majority of commercially available GDL materials are based on wet-laying technologies (carbon paper made from chopped fibers) since this technology features a sufficiently high productivity and the potential to achieve competitive costs. There are few systematic investigations of GDLs in the cathodic production of H_2O_2. The consortium partner **SGL Fuel Cell Components GmbH** is a **leading manufacturer of carbon and graphite materials** and will produce custom-made GDLs with tailored adjustments to maximize the three-phase boundary for maximum species conversion at the catalyst sites.

The anodic pathway (WOR) has gained considerable interest in the last years. WOR efficiency, selectivity, and stability has remarkably improved, especially due to catalyst and process development.[48] Several promising candidate materials for catalysts selective towards WOR have been identified, including oxides such as $BiVO_4$, $CaSnO_3$, and ZnO.[24-25] **Boron-doped crystalline diamond electrodes (BDD)** are nevertheless the most promising electrode material for highly efficient WOR.[26] BDD electrodes can suppress the most common side reactions in aqueous electrolytes: hydrogen and oxygen evolution in the cathode and anode, respectively. Figure 2 shows the BDD overpotential against hydrogen evolution reaction (HER, -1.2 V *vs.* SHE) and oxygen evolution reaction (OER, +2.5

[17] https://www.solvay.com/en/our-company/our-businesses/performance-chemicals/myh2o2-satellite-peroxide-production
[18] Xue, Yudong, et al., Electrochemical and photoelectrochemical water oxidation for hydrogen peroxide production. *Angew. Chem. Int. Ed.* **2021,** *60* (19), 10469-10480.
[19] https://ec.europa.eu/clima/eu-action/climate-strategies-targets/2030-climate-energy-framework_de
[20] Perry, S. C. et al., Electrochemical synthesis of hydrogen peroxide from water and oxygen. *Nat. Rev. Chem.* **2019**, 3 (7), 442-458.
[23] Traube, M. Über die elektrolytische Enstehung des Wasserstoffhyperoxyds an der Kathode. *Ber. Kgl. Akad. Wiss.* 2, (1887) 1041 [24]
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[21] Navarro et al., J. Chem. Soc., Faraday Trans. 1 (1984), 249.
[22] Yang et al., Toward the Decentralized Electrochemical Production of H_2O_2: A Focus on the Catalysis 2018, 4064-4081
[23] Bu et al., Carbon-Based Electrocatalysts for Efficient Hydrogen Peroxide Production, *Advanced Materials* **2021**, 2103266
[24] Siahrostami, S. et al., One- or Two-Electron Water Oxidation, Hydroxyl Radical, or H_2O_2 Evolution. *J Phys Chem Lett* **2017**, 8, 1157-1160
[25] Shi, X. et al., Understanding activity trends in electrochemical water oxidation to form hydrogen peroxide. *Nat Commun* **2017**, 8 (1), 701
[26] Cobb S. J. et al., Boron Doped Diamond: A Designer Electrode Material for the Twenty-First Century, *Annu. Rev. Anal. Chem.*, **2018**, 11, 463-484

V *vs*. SHE). The high anodic overpotential can generate OH^- radicals directly from water, which instantly oxidize the surrounding electrolyte. Such electrodes are already used industrially and can have a life cycle of several months to years in quite harsh electrolysis medium.[27]

[27] WO 01/25508 A1 Haenni et al. ELECTROCHEMICAL PRODUCTION OF PEROXOPYROSULPHURIC ACID USING DIAMOND COATED ELECTRODES

Because of their high overpotential for WOR, **BDDs lead to remarkable high efficiencies in anodic H_2O_2 production**. Through the design of advanced BDD materials, **FAU** has pushed outstanding progress in the efficiencies for H_2O_2 production from WOR,[28] with production rates of 74.6 µmol min^{-1} cm^{-2} and FEs of 87 % (peak maximum).[33] Besides its scientific importance, these reports involving catalysts or parameter screening are primarily carried out in a small scale (H-cell setup). Thus, POWER2HYPE will focus on the use of such robust electrodes in continuous flow cells and stacks setups, aiming the scale up in industry-relevant processes and applications of H_2O_2. Together with **DiaCcon**, a leading BDD producer, the technology transfer from lab to pilot scale can be reached within the POWER2HYPE timeline. More recently, **FHG** has shown a continuous flow process using a commercial BDD electrode in combination with an optimized carbonate-based electrolyte system and have achieved FEs up to 78 %, and H_2O_2 production rate of 79 µmol min^{-1} cm^{-2}, operating at current densities up to 700 mA cm^{-2}.[34]

Figure 2: Electrochemical potential window for HER and OER on different electrode materials (left), DiaCCon diamond coating machines facilities (middle) and Aquivion® polymer chain from Solvay (right).

POWER2HYPE will also address the **cell design** to achieve maximum H_2O_2 production. To produce H_2O_2 various cell configurations, undivided and divided by one or two membranes, can be used.[26] For water treatment applications, typically an undivided cell is used,[29,30] which can be related to the required low H_2O_2 concentration. However, to increase the H_2O_2 concentration, decomposition of H_2O_2 at the anode is to be avoided, which is normally done by dividing the cell by a proton exchange membrane.[22] In such a configuration a PEM fuel cell approach could continuously produce concentrated H_2O_2, however, concentrated H_2O_2 accelerates the cathodic reduction of H_2O_2 to H_2O.[26] **WR** demonstrated efficient H_2O_2 production utilizing a dual membrane configuration at an industrial relevant current density of 400 mA/cm^2 (Figure 3). Using this cell configuration the generated H_2O_2 is transported away from the cathode, thus avoiding high H_2O_2 concentrations near the cathode. Additional advantage is the alkaline environment at the cathode, in which the faradaic efficiency and productivity of (metal-free) carbon-based electrocatalysts are high.[22,26,27] The generated H_2O_2 is transported into a center compartment via electrodialytic principles resulting in a concentration of 11 % H_2O_2, with the anion exchange membranes (AEM) having a limited stability.[37] The incorporation of solid polymer electrolyte (SPE) in the center compartment discards the use of supporting electrolyte, resulting in the continuous production of pure H_2O_2 (about 7%, no salts, not additives) at high FE (>60%) for almost 100 h.[38] More recently, Xia *et al.* achieved 20 wt.% H_2O_2 using the same WR configuration, yet at a FE of ~20% and using hydrogen gas as additional feed.[31] A concentration of ~20 wt.% H_2O_2 or higher is in the range of the AO process before multiple purification steps and subsequent concentration to 50-70% by distillation.[17]

Figure 3: WR double-membrane setup for ORR (left and middle) and POWER2HYPE approach for paired electrolysis using a zero-gap cathode configuration (right).

Membranes have a crucial role in modern electrolytic processes. Proton Exchange Membrane (PEM) electrolysis is now growing in the electrolysis application for large scale green hydrogen production market (up to several GW

[28] Mavrikis, S. et al., Boron-Doped Diamond Electrocatalyst for Enhanced Anodic H_2O_2 Production. *ACS Appl. Energy Mater.* **2020,** *3* (4), 3169-3173.
[33] Mavrikis, S. et al., Effective Hydrogen Peroxide Production from Electrochemical Water Oxidation. *ACS Energy Lett.* **2021**, 2369-2377 [34] Pangotra, D. et al., Anodic generation of hydrogen peroxide in continuous flow. *Green Chem.* **2022**.
[29] Sopaj et al., Effect of cathode material on electro-Fenton process efficiency for electrocatalytic mineralization of the antibiotic sulfamethazine. *Chem. Eng. J.* **2020**, 123249.
[30] Pérez et al., Towards the scale up of a pressurized-jet microfluidic flow-through reactor for cost-effective electro-generation of H_2O_2, *J. of Cleaner Production* **2019**, 1259-1267.
[37] Kuehn, C. et al., The Electrolytic Synthesis of Hydrogen Peroxide in a Dual Membrane Cell. *J. Electrochem. Soc.* **1983** 130, 1117 [38] WO2017/222382 – Electrochemical Process and Reactor
[31] Xia, C. et al., Direct electrosynthesis of pure aqueous H_2O_2 solutions up to 20% by weight using a solid electrolyte. *Science* **2019,** *366* (6462), 226-231.

demonstrators today), as a new generation of electrolysers beyond alkaline electrolysis.[32] **Solvay** developed more than 20 years ago[33] its own PEM technology based on perfluorinated polymer backbone (Aquivion®, Figure 2),[34] to contribute to markets of proton exchange membrane such as redox-flow battery, electrolysis and fuel cell applications (Figure 2). A great advantage of this perfluorinated backbone is its super acid character, which offers very high proton conductivity and outstanding chemical durability.

Yet, membranes have an important role not only as ion exchange membranes (IEM) for separation of cathodic and anodic half-cells, as also as membrane electrode assemblies (**MEA**s). By pressing a GDL onto the anion exchange membrane (AEM), **WR** successfully demonstrated cathodic H_2O_2 electrosynthesis at 50 mA/cm^2.[35] Yet, the influence of MEA fabrication method and composition is unknown. In POWER2HYPE, the influence of the materials composition (membrane, catalyst, binder, ionomer) and MEA coating method (spraying, painting, etc) will be evaluated for achieving stable and long-term electrolytic performance.

As part of the MEA, the AEM is of great importance to transport formed perhydroxyl ions (HO_2^-) towards the center compartment (Figure 3). For the **improvement of the AEM,** POWER2HYPE aims membranes based on Solvay's Aquivion® proprietary perfluorinated backbone. Experiments of ORR with more stable commercial AEMs (Neosepta AHA and AMI-7001) resulted in a specific energy loss of 1.3-2.5 kWh/kg H_2O_2 (100%) attributed to their low conductivity. In contrast, Solvay's Aquivion® E87-05S (0.02 Ω·cm^2) results in a considerably lower specific energy loss of 0.01 kWh/kg H_2O_2. Solvay is a pioneer in AEM R&D[36] and new recent synthesis routes led to anionic exchange ionomers from Aquivion® with outstanding performances in ionic conductivity and durability.[37] In POWER2HYPE this anion exchange ionomer will be investigated with different charge carriers and side chains on the perfluorinated backbone to improve mechanical behavior and anion conductivity. Membrane formulation strategies will also be considered to improve membrane lifetime (membrane mechanical reinforcement and chemical stability) under alkaline and concentrated H_2O_2 conditions.

A commercial version of electrochemical ORR to produce H_2O_2 has already been developed by the Danish company HPNow. Besides its high purity, H_2O_2 is generated only in low concentrations (0.15%)[38] in small to medium sized devices, limiting application mainly to aquaculture and (waste)water treatment. In a currently running H2020 project PeroxyPro,[39][40] HPNow targets the application of the developed system in water treatment and disinfection. It is important to note, that these applications cover only less than 10 % of the current market for H_2O_2.[8]

Key approach of the POWER2HYPE process is to **pair the anodic and cathodic H_2O_2 generation in one cell** using non-critical and abundant materials only. This procedure adds substantial advantages compared to cells utilizing only one of the half-cells for H_2O_2 generation, where the second electrolyte cannot be valorized and is often discarded. **The ambition of the** POWER2HYPE **proposal is to progress to larger-scale H_2O_2 production targeting also applications requiring higher H_2O_2 concentrations,** which are to date not yet commercially addressed with electrochemical H_2O_2 production. Hence, POWER2HYPE focus specifically on applications where concentrations **of 3% and above** are required. The process performance will be enhanced by reducing the specific energy usage, thereby offering an **industrially relevant alternative for the AO process**. To realize our ambition, POWER2HYPE will build on recent progress made by the project consortium.

Table 3: Current research status of POWER2HYPE partners with KPIs for ambition.

Partner	Current research status	Ambition through
WR	Improved cell design for the continuous cathodic production of 'salt-free' 7% H_2O_2 at 64% FE for almost 100 h	Continuous cathodic production of at least 10 % H_2O_2 with 80% FE for 1000 h at ≤ 3 kWh/kg.
FHG	Anodic H_2O_2 production developed in continuous flow at lab-scale with the highest production rate reported to date.[48]	To improve the process with robust, scalable and economically viable BDD anodes, obtaining 3 wt% H_2O_2 continuously for 1000 h.
FAU	Most stable and efficient BDD anodes reported in the literature for WOR.	To produce stable and scalable BDD layers with economically viable electrode supports.

[32] Kumar, S. et al., Hydrogen production by PEM water electrolysis – A review. *Mater. Sci. Technol.* **2019**, 2 (3), 442-454
[33] Arcella, V.; Ghielmi, A.; Tommasi, G. Ann. N.Y. Acad. Sci. 2003, 984, 226;
[34] https://www.solvay.com/en/brands/aquivion-ion-conducting-polymers
[35] WO 2015/034354 – Electrochemical production of Hydrogen Peroxide
[36] WO2012/098146 – Liquid compositions of fluorinated anion exchange polymers
[37] Bonizzoni, S. et al., Facile Chemical Modification of Aquivion® Membranes for Anionic Fuel Cells. *ChemElectroChem* **2021**, 8 (12), 2231-2237.
[38] https://www.pennyukltd.com/tag/hpnow/, https://www.hpnow.eu/wp-content/uploads/2019/12/hpgen-almeria-case-study.pdf, https://www.rippleaquaplast.co.uk/wpcontent/uploads/hpnow-leaflet.pdf (visited 22-3-2022).
[39] https://cordis.europa.eu/project/id/960753
[40] Pangotra, D., et al., Anodic production of hydrogen peroxide using commercial carbon materials. *Appl. Catal. B Environ.* **2022**, 303, 120848.

DIA	DiaCCon (DIA) already produces electrolysers and BDD anodes, mostly for industrial applications.	To expand the portfolio with stacks for applications beside wastewater treatment and develop complex electrolysers using novel electrode and membrane setups
Solvay Group	Solvay as developed and offers cation-exchange membranes (CEM) with excellent lifetime for fuel cells, electrolyzers, and various electrochemical technologies (Aquivion®). It is the leading world supplier of H_2O_2.	To develop anion-exchange membranes (AEM) suitable for application in electrochemiAssociated wi document Ref. Ares(2022)7691164 - 08/11/2022cal processe and to enhance current product portfolio and t establish, demonstrate and apply a sustainable proces for H_2O_2 production.
WU	WU has vast experience carbon-based catalyst synthesis, characterization and performance. The use of renewable energy to convert sustainable resources is the focus area of application	To expand the capability of electrochemical conversions by using carbon catalysts/electrodes.
SGL	SGL offers various GDL materials for automotive, stationary and portable fuel cells and PEM electrolysers. Currently the key business lines are FCC (fuel cell components) in the business unit GS (graphite solutions).	To develop GDL materials tailored to the application in gas diffusion electrodes (GDEs) for cathodic H_2O_2 electrosynthesis. Fibrous carbon materials or graphite materials will be provided as substrates for deposition of BDD (anodes).
TRSA	TRSA has know-how in the cell and stack design, including fluid dynamics studies, the development of zinc-air batteries, as well as in the engineering of pilot plants around that technology.	TRSA will contribute to the optimization of the electrochemical cell and stack efficiency through research on the management of process fluids, assessment of the physicochemical properties of the utilized materials and simulation of the fluid dynamics
IRIS	IRIS is a deep-tech engineering company specialized in real time monitoring and industrial artificial intelligence solutions	Develop a digital twin aiming for real-time monitoring and control of the H_2O_2 electrochemical production process.
SLVG	Presently proven & patented concentration process in laboratory scale of 5 kg feedstock,[41] along with techniques for scaling up the chemical process, sweeping gas and membrane based. A supply chain has already been formed for sourcing of commercial components for research.	To demonstrate a large-scale concentration process capable of producing 20 kg concentrated H_2O_2, while maintaining a high process efficiency and gas saturation efficiency of over 80 %.

1.1.3 Technology maturity

Once key POWER2HYPE key components (currently at **TRL3/4**) validated at lab scale, they will be integrated in a pilot plant that will be demonstrated in relevant environment at Solvay's premises (Belgium) **(TRL6).**

Table 4: POWER2HYPE technologies and their rise in TRL over the project duration.

Technology	Rationale	Initial TRL	End TRL
Direct electrocatalytic reduction of oxygen and in-situ concentration of H_2O_2	Materials and cell design will be optimized to improve the energy efficiency, thus reduce the energy consumption, and improve the productivity, i.e., current density.	**TRL 3-4**: 7% H_2O_2 has been reached by WR in a 10 cm^2 single cell at a current density of 50 $mA \cdot cm^{-2}$ during 5 days of operation	After lab scale validation, as part of the paired electrolyser integrated in the pilot this technology will demonstrated in a system (**TRL 6**).

[41] WO2021096353A1 PROCESS AND APPARATUS FOR CONCENTRATING HYDROGEN PEROXIDE TO 98 WT.% OR MORE

Direct and continuous water oxidation to H_2O_2 on BDD anodes.	Materials, and cell design will be optimized in terms of energy efficiency. The electrolyte composition will be optimized in terms of energy efficiency and interconnection with down-stream processing and the cathodic process.	**TRL 3-4**: Successful tests at 400 mA·cm^{-2} in a 10 cm^{-2} flow cell for >2 days	Demonstration of the improvement by implementing the newly developed materials and electrolyte composition in a single cell in a laboratory environment. This technology will be demonstrated in a pilot system (**TRL 6**).
Combined cathodic and anodic H_2O_2 production (paired electrolysis)	The cathodic and anodic H_2O_2 are integrated in one cell and the paired reactions validated (TRL 3) and subsequently demonstrated on lab-scale (TRL 4)	**TRL 3**: A new concept of paired electrolyzer is developed. The coupling in TRL3 has already been done at FHG.	Demonstration and validation of the paired electrolysis in a single cell at lab scale. This paired electrolyser will be upscaled and demonstrated in a pilot plant system (**TRL 6**).

Electrolyser stacks comprising of a multitude of cell pairs	Design, construction and build of a paired electrolyzer with cell pairs of each 200 cm^2 to produce up to 20 kg per day (10 kg per day by anodic and 10 kg per day by cathodic process);	**TRL 4**: GDLs and BDD electrodes can already be manufactured at 200 cm^2 by SGL and DiaCCon. But at this stage only electrolyser cells from 10 cm^2 single cells have been tested	The 200 cm electrolyser stack will be integrated in a system, where its operation will be validated (**TRL 5**) and demonstrated over 1000 hours in a demo plant at Solvay to produce up to 20 kg H_2O_2/day (**TRL 6**).
Design, integration, and optimization of a DSP to obtain H_2O_2 at any given concentration.	Scaling guidelines for DSP will be devised in order achieve an energy efficient concentrating equipment and to ensure capability of concentrating 200 kg feedstock. The influence of the dynamic production, due to the intermittency of renewable energy, is considered. The surrounding sub-systems will also be sized.	**TRL 3**: Concentration proven in laboratory pilot scale with feedstock of 5kg, with chemical analysis completed	Integrated in the demo plant at Solvay, DSP of H_2O_2 and its flexibility in terms of user controlled output H_2O_2 concentration is demonstrated (**TRL6**)

Associated with document Ref. Ares(2022)7691164 - 8/11/2022

1.2 Methodology

1.2.1 Concepts and assumptions behind POWER2HYPE

While the electrochemical chlorine production is a well-established technology in industry and hydrogen production is already available at pilot scale, electrochemical production routes for other chemicals are currently limited to low TRLs, many times with low efficiencies and low production volumes, and mostly restricted to niche applications. **In** POWER2HYPE **the target is to develop an electrochemical technology to produce hydrogen peroxide, an important bulk chemical, essential for a wide range of applications. In combination with renewable electric energy sources, it aims to set another milestone towards the low-carbon electrification of the process industry.** Electrochemical H_2O_2 production meets all requirements for a "green", sustainable production process: the required educts are air (oxygen), water, and electricity. Both electrodes of the paired electrolyser generate the target product, and no by-products are generated besides low amounts of O_2, H_2O and H_2. Neither the anode nor the cathode requires any critical or toxic raw materials. The electrolyser is designed for operation at high current densities of at least 400 mA cm^{-2} to enable high production rates per electrode area. Nevertheless, the electrolytic process with DSP (and downstream derived bulk chemicals) should be flexible to run under part-load or in intermittent operation (switchon and –off within few minutes and mainly determined by the periphery), e.g., when electricity from renewable resources should not be sufficiently available.

Climate change targets and European dependence on fossil-based energies
The main objective of the European Green Deal is to achieve carbon neutrality in the EU by 2050. As an important milestone towards this goal, the EU's net greenhouse gas emissions should decrease by at least 55% by 2030. However, as energy prices have dramatically risen due to changes in the geopolitical landscape, the European Commission (EC) proposed on the 8th of March 2022 an outline of a plan to make Europe independent from Russian fossil fuels well before 2030, starting with gas. This European initiative for more affordable, secure, and sustainable energy (**REPowerEU**)[42] seeks, among other measures, to rapidly adopt technologies allowing for replacing gas in power generation. Such joint effort can reduce the European demand for Russian gas by twothirds before the end of the year 2022. One of the pillars of the REPowerEU initiative is the fast reduction of fossil fuels utilization in the industry and power system by **a) boosting energy efficiency**, b) **increasing the share of renewables** in the energy mix and c) **enhancing electrification of industrial processes**.
The moment for renewable energies in the industry

[42] REPowerEU: Joint European action for more affordable, secure and sustainable energy

Future power generation should include a renewable share of 68% in 2030 and 86% in 2040 (Figure 4).[43] The direct connection of industrial production plants to intermittent renewable energy sources (e.g., PV) has emerged as a competitive and cost-effective solution to enable the electrification of chemical processes. Thereby, the chemical industry can strongly benefit from an already enhanced decarbonization of the power sector. The tremendous rise of renewable power generation and the associated drop in production cost, reached costcompetitiveness with conventional electricity generation.[44] The costcompetitive generation of renewable electricity has opened new possibilities of "electrifying" production processes and given new momentum to the develop-ment and application of Power-to-X processes in the chemical industry, i.e., to the "electrification of the chemical industry". In this context, an important technical requirement arises for technologies and processes to be applied for this "electrification": As mentioned above, renewable energy sources are typically characterized by a certain intermittency. Thus, new processes and process technologies need to provide a certain load flexibility, being able to be safely and stably operated at different load factors, while still providing sufficient performance when running outside of the optimal parameter window. This is an important difference to conventional processes, typically designed for very constant supply of fossil feedstock. Load flexibility is also an important process parameter to be addressed in POWER2HYPE (WP6 Demonstration).

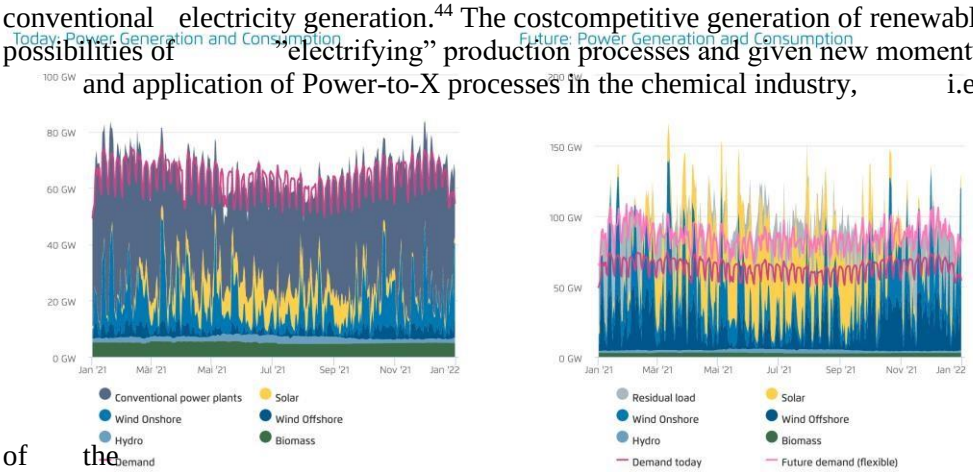

Figure 4: Power generation and consumption for 2021 (left) and 2040 (right). 2030 should achieve a renewable share of 68%, and in 2040 of 86%. Source: Agorameter.

1.2.2 Overall methodology

The 48-month **POWER2HYPE** project is divided into a set of distinct, interlinked R&D activities running in parallel or in sequence (Figure 5).
The main R&D work (WP2-6, Figure 6) will start by investigating the catalysts and materials in WP2 (anode, cathode, GDL and membranes) and its performance in each halfcell (cathode & anode). The materials will be used for development of the electrochemical process in WP3.

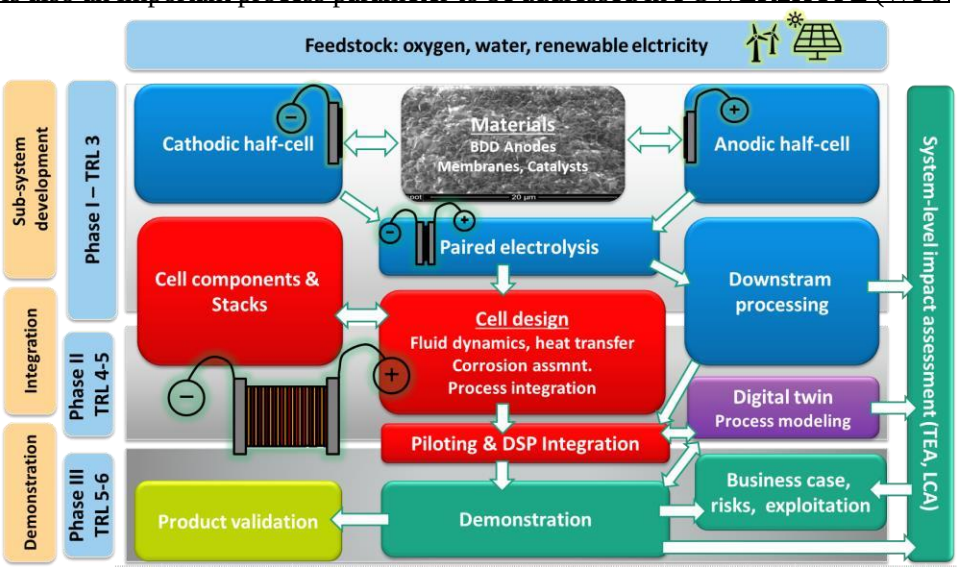

Figure 5: Overall work breakdown structure of POWER2HYPE indicating 3 interlinked project phases over which the TRL levels are raise from 3 to 5-6.

The processes will be customized and optimized to be efficiently paired into one electrochemical cell using 10 cm^2 cells (TRL3).

[43] www.agora-energiewende.de
[44] IRENA Renewable Cost Database (https://www.irena.org/)

The paired process will be scaled up in to 200 cm^2 cells (TRL4) in WP4. The cells will be stacked (TRL5) and integrated with a DSP process. The integrated process will be engineered (WP5) and demonstrated (TRL6) in WP6 with renewable (and intermittent) energy supply.

Moreover, impact assessments in WP7 will investigate the larger environmental, societal and economic impacts of POWER2HYPE. WP7 keeps monitoring the results of WP2-6 during the three phases of the project, helping to set and adjust project performance goals. Critical output of the System Level Impact Assessment is the comparative analysis showing the potential strengths of POWER2HYPE and the milestones this technology must reach to be ultimately economically competitive, environmental beneficial and socioeconomically advantageous. Associated with document Ref. Ares(2022)7691164 - 08/11/2022 In addition to that, non-technical activities will be running continuously, such as project management, exploitation, communication, and dissemination.

Phase I – M1-24, TRL 3

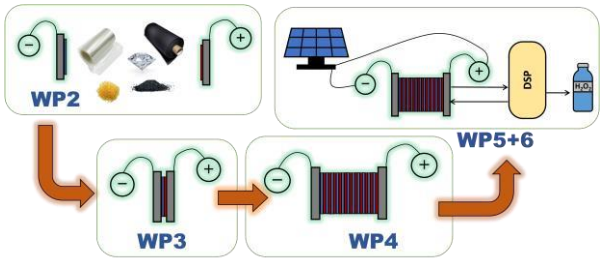

Figure 6: Electrochemical activities from WP2 to WP6.

The first phase of the POWER2HYPE project will address the fundamental aspects of the electrochemical H_2O_2 production. **SGL, FAU, WU, WR and RHDO** will work on the design, synthesis, and characterization of advanced component materials, such as cathode catalysts, gas diffusion layers/gas diffusion electrodes (GDL/GDE), BDD electrodes, and cation and anion exchange membranes (CEM/AEM). The focus will be on: (i) the efficiency and selectivity of materials (suppressing competing reactions like H_2O_2 electro-decomposition, H_2- and O_2-evolution); (ii) stability under alkaline conditions and H_2O_2 concentrations of 20 % (especially challenging for AEM), and (iii) avoiding critical or toxic materials by using mainly carbon-based. Regarding membrane development, Solvay will develop new alternatives ionomer solvents (perfluorinated free) for CEM and will build upon Aquivion® for AEM.

The tailored materials will be provided (and when needed adapted) for the development of an efficient electrolytic process for the H_2O_2 generation (**FHG, WR** and **SSPI**). **WR** will develop the cathodic oxygen reduction reaction (ORR), whereas **FHG** will work in the anodic water oxidation reaction (WOR). Both will use cation and anion exchange membranes (CEM and AEM) developed and provided by **SSPI**. Work on ORR will emphasize on an electrolyte-free process using catalyst coated GDE or a membrane electrode assembly (MEA) configuration to obtain pure H_2O_2 (no salts, no stabilizers) with concentration of at least 10 wt.%. Research on the anode will be focused towards suppressing the oxygen evolution to favor the H_2O_2 formation from water using BDD electrodes. Strong focus is placed on a swift transfer from discovery and screening toward deployment and full single electrolyser-cell design/testing of cathode/anode catalyst formulations in small-scale of 10 cm^2 cells. The ORR and WOR processes will be paired with suitable electrodes and electrolytes and the TRL3 electrolyser will be used as basis for scaled up electrolysis carried out in Phase 2 and 3.

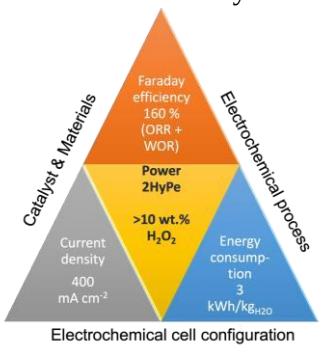

Figure 7: POWER2HYPE KPIs.

Finally, within this phase I, a lab-scale cell (10-25 cm^2) will be designed and manufactured using input from the previous experimental findings. Corrosion tests for components resistance to H_2O_2 will be carried out by **DIA**. Cell tests will be carried out by **FHG**, including lab-scaled paired electrolysis.

At the end of the first phase, **the decision milestone 1 will define the status of the project** (explained in more details in 3.1 Work Plan). This decision milestone takes into account a process in lab scale with efficient catalysts (160 % total FE) at a current density of 400 mA/cm^2 consuming less than 3 kWh/kg_{H2O2} as shown in the target triangle of POWER2HYPE (Figure 7).

Phase II – M24-36, TRL 4-5

With the laid foundation for the paired electrolysis working in lab-scale, **Phase II of POWER2HYPE will focus on scaling up the electrochemical process and integration with DSP**.

Important activities will include the **scale up of electrolyser cells from 10-25 cm^2 to 200-400 cm^2** (DiaCCon, SGL) using input from experimental findings obtained from preceding tasks. DiaCCon, FHG, TRSA and SGL will design an electrolysis stack prototype containing feed channels for gas supply/outlet and electrolyte supply/outlet as found appropriate for the electrodes and materials. The **stack** will be designed to achieve a **H_2O_2 production rate of up to 20 kg/d (based on 100 wt.% H_2O_2)**, corresponding to an **electrical power of approximately 2.5 kW**. The design, which will be supported by TRSA's and FHG´s simulations on fluid flow and heat management, will involve the

scale-up of the electrodes (to 200 or 400 cm^2 per electrode), development of machined graphite bipolar plates (BPP) with flow fields on GDL side, and the unfolding of the most suitable stack configuration (monopolar or bipolar electrodes, as shown in Figure 8). All supporting infrastructure to ensure both smooth and safe operation of the system will be considered and implemented as part of the entire hardware assembly.

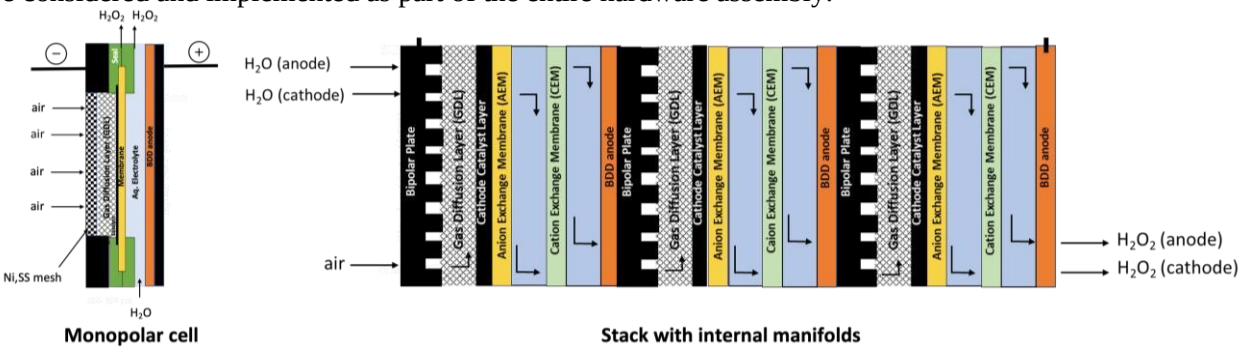

Figure 8: Powe2Hype design for a zero-gap electrochemical monopolar cell and stack with internal manifolds.

Another important activity will be the development of a **low-energy down-stream process (DSP) by SolvGe**. The DSP consists of a device for concentrating the 3% and 10% H_2O_2 streams using the vapor pressure difference between hydrogen peroxide and water molecules. This is achieved by imparting a velocity differential between a gas and the liquid medium to promote a high mass transfer rate. Optimal mass transfer rates allow the process to occur at ambient conditions. The concentration reactor will be designed to ensure a high sweeping gas saturation rate, leading to a low energy consumption of the DSP. The properties of the sweeping gas can be varied to control the concentration rate and maximum output concentration, in a completely automated and flexible manner.

TRSA, IRIS, SolvGe and Solvay will focus on the design, integration and testing of the electrolyzer and DSP system with the necessary auxiliary systems, fluid management, control and safety, electricity supply and management to pave the way for the demonstration of POWER2HYPE pilot plant in Phase III. A digital twin of the demonstration will be designed by **IRIS**, and the process chain will operate virtually before the physical prototype, thereby finding important bottlenecks in the process. The **H_2O_2 production will be integrated with the DSP** to supply a single pilot unit that can be completely operated from one control panel.

At the end of the second phase, **the decision milestone 8 will define the status of the project**. With this decision milestone it will be evaluated if the stacks can produce H_2O_2 in the targeted amounts (20 kg/d, 3 kWh/kg H_2O_2, 160 % FE, 400 mA cm^2) and the integration with DSP is accomplished.

<u>Phase III – M36-48, TRL 5-6</u>

The last phase of POWER2HYPE will concentrate all efforts into the demonstration of the reaction and process that turns water, air and renewable electricity into the production of H_2O_2. **This activity is covered by Solvay and TRSA and represents the highlight of the POWER2HYPE project**. The objective is the demonstration at **TRL 5-6** of a viable and integrated electrochemical process in an industrial environment with the capacity of producing **20 kg/day (based on 100 wt.% H_2O_2)**. Although there is the potential to concentrate H_2O_2 up to 99 wt.%, the DSP will tune for a maximum of 70 wt% for storage-safety reasons. The whole process chain is to operate at efficiencies meeting and exceeding those of competing processes. Furthermore, in the European-wide roll-out of the technology, its connected economic, environmental and societal impacts will be assessed by EI, TRSA, IRIS and Solvay. IRIS will transfer the technical data into a virtual demonstration for extensive optimization modeling and predictive robust process integration and upscale. The time-resolved upscale approach provides the ability to follow the dynamics of fluctuating renewable power supply to be taken into account, so that a GHG-neutral process can also be ensured in operation. Technological learning curves considering component cost level will be designed based on literature data and POWER2HYPE results. This learning curve model will serve as input for an early TRL CAPEX, OPEX and economic viability assessment, including sensitivity analyses.

1.2.3 National and international research and innovation activities linked to POWER2HYPE

The POWER2HYPE project plans to implement networking activities and links with existing projects in order to consider and compare the research and results implemented and include when relevant their results into the project. This task will be facilitated by the fact that several partners in the project are already involved in initiatives using similar technologies, and closely related to the topic of electrochemistry and sustainable processes.

*Table 5: Related past and current projects of **POWER2HYPE** consortium partners where outputs feed into the project.*

Initiative / project	Description	Partner

CO2EXIDE – (H2020)	CO2-based electro synthesis of ethylene oxide Process development for the anodic H_2O_2	FHG, EI
LIBERATE – (H2020)	Lignin biorefinery approach using electrochemical flow	FHG
"**Electricity as a resource**" – FHG Lighthouse intern project	Development of electrochemical process, components, and system evaluation	FHG
FP7 Necobaut	Develop a new concept of battery for automotive based on a new metal/air technology that overcomes the energy density	TRSA
	limitation of the Li-ion batteryAssociated with document Ref. Ares(2022)76	
EU Life Zaess	Development and demonstration of an innovative zinc-air energy storage technology that can address the challenges of integrating intermittent renewable energy. This innovative zinc-air flow battery combines the advantages of metal-air batteries and flow batteries.	TRSA
EEA GRANTS SAMER	Development of a proprietary zinc-air flow battery technology for grid scale storage of renewable energy. Zinc-air flow batteries are a low cost, safe and environmentally friendly energy storage technology.	TRSA
Green Oxidants (National funding)	Decentralized production of H_2O_2 focussing on the 2 electron ORR solely.	WR
Hyper (Dutch science foundation (DSF)	Project on the electrocatalytic H_2O_2 production via the 2 electron ORR using noble metal free catalysts and new membranes.	WU, WR, Solvay
Versatile and sustainable energy sources for society (DSF)	Proof of concept and development of a H_2O_2 concentration system, from laboratory to small pilot scale.	SolvGE

1.2.4 Interdisciplinarity of the project

In POWER2HYPE all the partners have knowledge and expertise revolving around the electro-chemical sector and cover different links of the H_2O_2 value chain (Figure 9). The consortium is composed by solid, experienced and leading industrial organizations, which possess substantial resources and networks to follow through with the project. The consortium is also counting renowned universities, research institutes and sophisticated technology developers representing the various areas of expertise that contribute to the design and development of the proposed solution with all entities coming from countries all across Europe. Finally, dedicated partner will perform communication to publicity, dissemination and exploitation to stakeholders.

Figure 9: The POWER2HYPE consortium.

1.2.5 Social science and humanities

POWER2HYPE contributes to Social Sciences and Humanities (SSH) dimension in three major socio-economic aspects, namely **competitiveness, climate change and energy dependency**. As the project enables a new sustainable electrochemical process for H_2O_2 production, it contributes to decarbonization of the industrial sector by changing an established chemical route. Specifically, the new electrochemical process will achieve a significant reduction in the overall energy efficiency compared to the classical anthraquinone (AO) process (Table 1). In addition, because the design makes use of sustainable and abundant materials as catalysts and cell components, it becomes inherently less burdensome from a resource demand standpoint, which simultaneously contributes to regional strategic economic independence and sector competitiveness.

Furthermore, concerning the actual material and energy demands of the POWER2HYPE process, no waste/side product will be generated (side reactions lead to low amounts of O_2, H_2O and H_2 only). In another words, the new process will transform this industry into one of **Zero-Waste Manufacturing**. The new process will also enable **a low-energy downstream process** to provide pure H_2O_2 at any desired concentration up to 99 wt.%.

Finally, it will improve the dynamics/agility of the process by allowing it to adapt and fully rely on the intermittent availability of renewable electricity. Consequently, an additional and very significant advantage will be to enable the **decentralized H_2O_2 production**.

1.2.6 Gender dimension

POWER2HYPE includes the gender dimension at various levels, as it is a transversal matter. The gender, diversity and equality dimension will play a role in the socio-technical analysis in WP7, i.e., in the stakeholder assessment, taking in consideration how the gender dimension can be of importance in risk perception or acceptance of a technology. Also, the roll-out/commercial setting of a technology potentially influences the attractiveness to people with e.g. different genders or family statuses, as e.g. the location or operational mode of a plant directly influences to whom a working position is attractive. All the partners in the project have gender equality policies that will be put forward and complied with during the project. For instance, **Solvay** has stated its commitment to be a completely gender-balanced organization by 2030. **EI** is part of the Equal by 30 initiative[45] and has a gender, diversity and equality plan implemented. The **FHG** has a goal to increase the proportion of female workers in research and ensure the reconsiliation of work, private and family life. The POWER2HYPE consortium commits to promotion of gender equality throughout its implementation cycle. Specifically, the participation of female scientists in managerial and technical coordination responsibilities in the project will be encouraged, thus pushing their careers. The following female scientists will lead tasks at various levels in the POWER2HYPE project:

*Table 6: Gender dimension in **POWER2HYPE**.*

FHG	Dr. Luciana Vieira (project coordinator), Lisa Wünsche / Jenny Ulrich (administrative).
SolvGE	Dr. Jyoti Botchu Vara Siva (CEO) and Larissa Arif (product engineer)
EI	MSc. Valerie Rodin performs (TEA & LCA), Dr. Karin Fazeni Fraisl (LCA and sociotechnical analysis), Dr. Christina Friedl (stakeholder, sociotechnical and STEEP analysis).
IRIS	Ana Maria Lopez and Alexandra Poch are project & innovation managers
WU	Dr. Guanna Li (team scientist) and Susan Witte (technician)
Dia	Dr. Hanadi Ghanem (team engineer for chemistry)

Furthermore, most consortium members support the principles as set out in the European Charter for Researchers & the Code of Conduct for their recruitment and are dedicated to responsibly comply with the principles of nondiscrimination, gender balance and the provision of fair framework conditions to all researchers.

1.2.7 Open science strategy

Open science practices (i.e. an approach based on open cooperative work and systematic sharing of knowledge and tools as early and widely as possible in the process) will be implemented as an integral part of our proposed methodology and for increasing our chances of delivering our objectives, as follows:

- **Early and open sharing of research:** publications in open access journals ensuring early sharing of publications (pre-prints) to satisfy open access requirements and enabling quick sharing of knowledge via deposit in repositories. Participation in open peer-review will also be encouraged to ensure transparency and support the shift towards Open Science.
- **Open access to research data and other related research outputs** (e.g. to software, algorithms) will be promoted following the principle 'as open as possible, as closed as necessary' and our strategy is to ensure that Open Science and IPR Management and closely linked.
- **Research output management:** all research outputs will be documented by the generators (researchers) of the outputs. Aspects as ownership, novelty, IPR protection, etc. will be established in a timely manner, and knowledge, data and information available for open dissemination will be confirmed prior to publication.
- **Measures to ensure reproducibility of research outputs:** relevant technical standards will be adhered to in order to ensure that scientific information, publications, data and other outputs from our project, as well as the metadata about them, is available for re-use in the long term. This will involve the use of persistent and unique identifiers, of certified repositories that are compliant to the standards of the European Open Science Cloud and in compliance with the FAIR principles for the management of research data produced by projects.

[45] https://www.equalby30.org/en

1.2.8 Data management

The POWER2HYPE consortium will deal with different type of data during the project activities: (i) scientific data coming from the project research activity and derived by merging, elaborating and analyzing existing data and (ii) data from press, blogs, forums and social networks. **Scientific data**

Regarding the intellectual property management and protection of results, the consortium will examine the possibility of protecting the project results (especially those expected to be commercially or industrially exploited) according to the Grant Agreement. In order to monitor this during the project, the consortium has appointed **FHG as Innovation and IP manager**. The following depicts the preliminary IP strategy of POWER2HYPE:

- Research and technology development partners retain the IPR of the technologies they invent and benefit from their eventual industrial uptake as well as expand their scientific excellence.
- Technology developers expand their application portfolio and retain the right to commercialize new application in cooperation with industries (research and technology development).
- The end-users are granted priority in applying the novel technologies in cooperation with POWER2HYPE technology providers and the research and technological development partners

Background IP and outputs will be identified and agreed by project partners within the Consortium Agreement (CA)

on base of the common and approved DESCA model. In general, the Consortium is committed in making all research data findable, accessible, interoperable, and reusable (FAIR), in compliance with the "Guidelines on FAIR Data Management in Horizon Europe". Data will be periodically stored in reliable facilities that permit recovery and do not compromise their integrity European Open Science Cloud (EOSC).

Data from press, blog, forum, and social network

Press articles are public and are not associated with any ethical issues. For what concerns data coming from blogs, forums, and social networks, POWER2HYPE will comply with the privacy policy signed by the users for writing information on the blog, forum or social network.

Data Management Plan

POWER2HYPE will collect and analyze data collected during the project and data that are publicly available or for which agreements will be established during the first months of the project. Data will be formalized in (semi)structured databases for their analysis in the project. Data use, if necessary, will be granted under a specific non-disclosure agreement. Data collected will never include information raising ethical issues (e.g., personal). Additional cases that may arise will be regulated according to the Grant Agreement and the Consortium Agreement. POWER2HYPE recognizes the value of regulating research data management issues and foresees the delivery of a Data Management Plan (DMP). The DMP will define how research data will be handled during the POWER2HYPE lifetime and after it is completed. It will be a dynamic document that will be delivered under the leadership of the coordinator and with a strong involvement of the General Assembly.

How will this data be exploited and/or shared/made accessible for verification and re-use?

All data will be first made anonymous to remove personal details, and then published in the EOSC. The POWER2HYPE consortium will guarantee the open access to its publications according to the Grant Agreement. It will undertake the following measures: depositing the project publications in a repository selected from the recommended lists of online archives provided by OpenAIRE (http://www.openaire.eu); providing Open Access publishing ("Gold") to the project publications; ensuring open access to the bibliographic metadata through the identification of the deposited publication via the chosen repository, and; Creative Commons License to the Online Platform / Website under an Attribution-Non-Commercial 4.0 International (CC BY-NC 4.0). **How will this data be curated and preserved?**

Every partner will be responsible for the secure storage of the partner's own generated data in reliable facilities that permit recovery of lost data and that do not compromise data integrity, such as the intranet of the POWER2HYPE website and OpenAIRE compliant data repositories (once results are published).

2. Impact

While the EU aims to become to become the first major economy to become "climate-neutral" by 2050, the chemical industry remains an energy intensive industry (612 TWh in fuel and power consumption in 2014, sector-wide, with 120 Mio tons CO_2 in related emissions). A large amount of manufactured chemical products are supplied to virtually all downstream sectors, which range from rubber and plastics, construction, automotive, pulp and paper, textiles to agriculture.[46] For many processes in the chemical industry and other sectors, defossilization is intrinsically complex considering the use of fossil feedstocks and the challenge to switch to renewable carbon-based feedstocks. Alternative chemical pathways to non-carbonous chemicals are in that sense less complex and can play a frontrunner role in the transition to more sustainable production pathways through "electrification" of their production processes. To significantly contribute to the transition towards a carbon-neutral economy with the milestone of reaching 60% emissions reduction by 2040 and the final target of full carbon-neutrality by 2050, it is vital to realize a technology breakthrough with the focus on bulk chemicals. H_2O_2 is a bulk chemical, with a production capacity in Europe of 1.2 million tons,[8] leading to significant GHG emissions of 2.4 million tons CO_2,[13] which is 1.8%, to the total GHG of the chemical industry in Europe.[14] Electrification of the current H_2O_2 process is therefore an ideal case

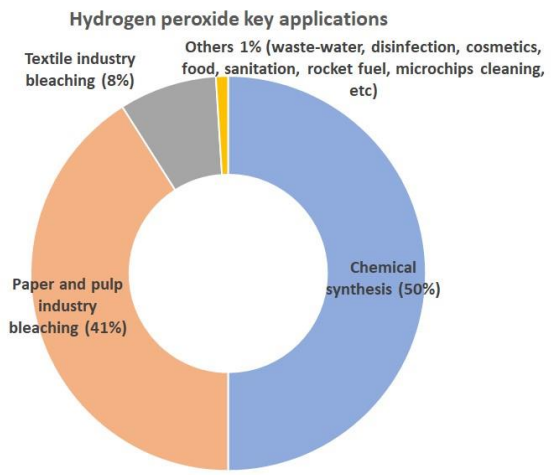

Figure 10: Hydrogen peroxide key applications.[8]

[46] https://cefic.org/a-solution-provider-for-sustainability/a-journey-to-sustainability/low-carbon-energy-and-feedstock-for-the-european-chemical-industrystudy/

for the storage of renewable electrical energy in chemical bonds and thereby creating a huge impact given the significant GHG emission reduction potential of the chemical industry in Europe and beyond.

Hydrogen peroxide production can be a first-of-a-kind demonstration of a new electrochemical process facilitating a more sustainable future for the chemical sector. The switch from the AO process to POWER2HYPE technology will have an impact on most abovementioned sectors from a life cycle perspective and will also bring benefits to several key H_2O_2 applications (Figure 10) thanks to technology transfer.
If POWER2HYPE is successful, the technology can reach TRL 9 by 2035 and, considering Solvay's customer portfolio, it has the potential to take a considerable part replacing its **40% of the European predicted production capacity for H_2O_2**. Following this implementation trajectory, by 2050 there could be a complete switch of the production model for hydrogen peroxide, as illustrated in Figure 11.

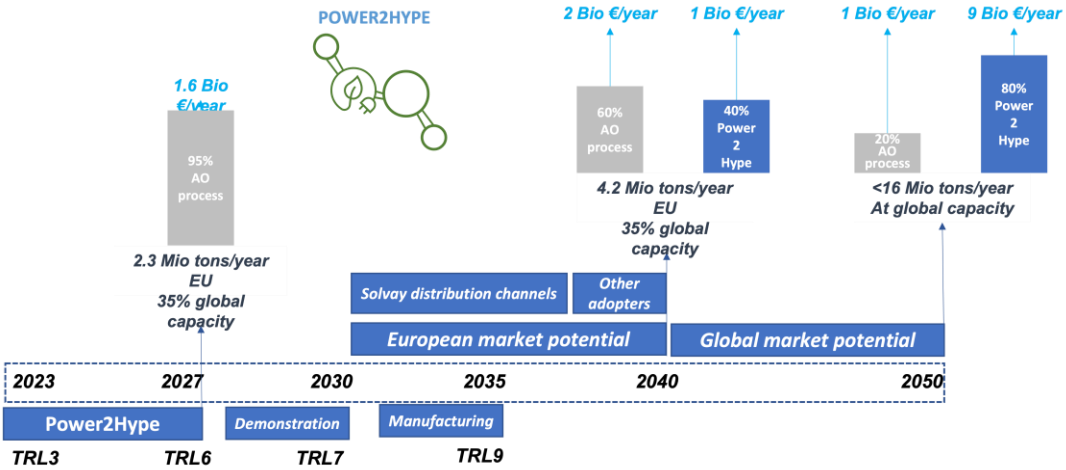

*Figure 11: Implementation roadmap for **POWER2HYPE** technology. Following full-scale demonstration of the technology after **POWER2HYPE**'s end and reaching TRL9 by 2035, market uptake is projected to reach a share of 40% in Europe by about 2040, corresponding to a market volume of 1 billion Euro, and to 80% or more of the global market by 2050.*

2.1 Expected impacts

2.1.1 Project outcomes and contribution to key strategic orientations (KSOs) and expected impacts

Outcome 1: Electrification of the industrial production process by shifting from the chemical conversion process to an electrochemical conversion process
State of play: Anthraquinone oxidation (AO) process: multi-step chemo-catalytic process requiring Pd-based catalysts, organic solvents, hydrogen gas, and energy input. Hydrogen gas is still produced mainly by steam reforming of natural gas (CH_4). Concentration of raw H_2O_2 takes place by distillation. POWER2HYPE **contribution:** The multi-step AO process is replaced by direct electrochemical production of H_2O_2via paired electrolysis of water and oxygen from air. POWER2HYPE will demonstrate this electrochemical production on pilot-scale and identify possible implementation sites of the process. **Main beneficiaries/stakeholders:** i) end-users will have a "green" alternative for conventional H_2O_2 supply; ii) operators of renewable energy plants who can produce chemicals in periods of low energy demand; iii) chemical parks with adjacent renewable energy plants; iv) civil and political society through a new technology contributing to a climate-neutral economy and limiting global warming.
KPI (associated): Successful demonstration of "electrifying" H_2O_2 production: POWER2HYPE pilot plant is successfully operated at Solvay, producing up to **20 kg/d of H_2O_2**.
Contribution to KSO: C and A **Contribution to expected impact in cluster 4:** sustainable by design advanced technologies **Contribution to specific objectives of the Processes4Planet:** S.O. 1 Integrating renewable energy
Outcome 2: Efficient integration of renewable electricity to drive the conversion process

State of play: The energy demand of the AO process is covered by a mix of fossil and renewable energy sources, varying from plant to plant. The hydrogen gas required for the AO process is produced mainly by steam reforming of natural gas.
Project contribution: The POWER2HYPE project will demonstrate highly efficient electrochemical H_2O_2 production (3 kWh/kg_{H2O2} at a current density of 400 mA/cm^2) using renewable energy sources only. In comparison, an AO process fed with hydrogen gas from electrolysis could cover only 75% of its energy demand by electricity from renewable resources (3 kWh/kg_{H2O2} out of approx. 4.4 kWh/kg_{H2O2}). To take advantage of intermittency of renewable energy, H_2O_2 production will be maximized at a surplus availability of electrical energy
(low costs) and minimized at a high demand of electrical energy (high costs). As such, the electrochemical process

acts as a buffer to store renewable energy in chemical bonds. This Associated with document Ref. Ares(2022)7691164 - 08/11/2022coupling with the energy sector envisioned to

take place at large scale, as similarly already done with the chlorine-alkali process, but as well as on a decentralized scale making use of an efficient smart grid integration.
Main beneficiaries/stakeholders: i) chemical and energy industry through sector coupling; ii) H_2O_2 end-users with ambitions to minimize their climate footprint; iii) H_2O_2 producers being able to offer climate-friendly production technologies; iv) civil and political society through a new technology contributing to a climate-neutral economy and limiting global warming.

KPI (associated): Electrochemical H_2O_2 produced at **400 mA/cm^2, FE of at least 160%** (both half-cells) and specific energy consumption of **3 kWh/kg_{H2O2}** or less for the electrolytic step, completely driven by renewable electricity.

Contribution to KSO: C and A
Contribution to expected impact in cluster 4: sustainable by design advanced technologies
Contribution to specific objectives of the Processes4Planet: S.O. 1 Integrating renewable energy

Outcome 3: Significant reduction of CO_2 emissions of the overall industrial process, including the emissions related to the generation of the electricity

State of play: Indirect CO_2 emissions of the AO process are estimated in 2.49 kg CO_2 per kg H_2O_2 produced, corresponding to current annual CO_2 emissions of 3 Mt per year in Europe.
Project contribution: The use of only water, air and renewable electricity as sole feedstocks in the POWER2HYPE technology will decrease the CO_2 emission of the electrochemical production process to close to zero. CO_2 emissions associated with H_2O_2 purification and concentration will be kept at a minimum by implementation of a novel concentration technology, also driven by renewable energy.
Main beneficiaries/stakeholders: i) H_2O_2 end-users with ambitions to minimize their climate footprint; ii) H_2O_2 producers being able to offer climate-friendly production technologies; iii) civil and political society through a new technology contributing to a climate-neutral economy and limiting global warming.

KPI (associated): Reduction of the CO_2 emission per kg of H_2O_2 in the complete value chain (including concentration) **by 90%** vs. conventional AO process, i.e. from 2.49 down to <0.25 kg CO_2 per kg H_2O_2.

Contribution to KSO: C and A
Contribution to expected impact in cluster 4: Clean and climate neutral value chains
Contribution to specific objectives of the Processes4Planet: S.O. 3 Moving towards commercially viable climate neutral and circular industry solutions

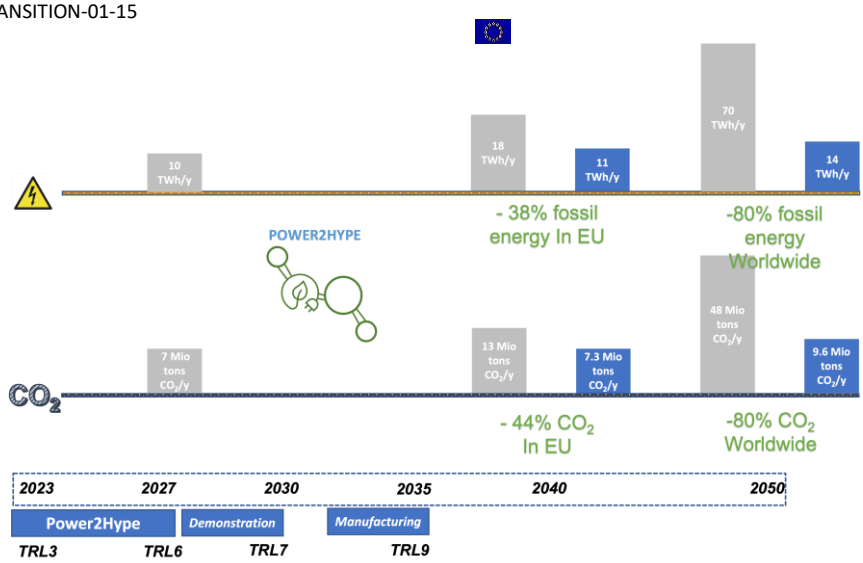

Figure 12: POWER2HYPE's impact roadmap in terms of electricity consumption and CO_2-related emissions. Roll-out of the POWER2HYPE technology could result in 38% less fossil energy consumption and 44% less CO_2 emissions in the EU by 2040 and in 80% less fossil energy use and CO_2 emissions worldwide by 2050, compared to the conventional business-as-usual H_2O_2 production scenario.

Outcome 4: Energy savings compared to the classical production routes
State of play: The anthraquinone process consumes 16×10^6 J/kg = 4.4 kWh/kg, even though the overall process ($H_2 + O_2 \rightarrow H_2O_2$) is thermodynamically favorable and theoretically requires no additional energy input ($\Delta G°_f = -2.6\times106$ J kg^{-1}, $\Delta H°_f = -5.5\times106$ J kg^{-1}). Thus, reducing the energy demand and decreasing the ecological footprint in the production of H_2O_2 would contribute to sustainable global development. **Project contribution:** POWER2HYPE target is an electric energy consumption of 3 kWh/kg for the electrosynthesis step, which means a reduction by more than 30% compared to the classical AO process. **Main beneficiaries/stakeholders:** H_2O_2 producers (efficient production)

KPI (associated): Energy saving of the POWER2HYPE process compared to the AO process, demonstrated pilot scale (target: **30% reduction**) Associated with document Ref. Ares(2022)7691164 - 08/11/2022
Contribution to KSO: C and A **Contribution to expected impact in cluster 4:** Clean and climate neutral value chains **Contribution to specific objectives of the Processes4Planet:** S.O. 3 Moving towards commercially viable climate neutral and circular industry solutions
Outcome 5: Overall material savings (waste reduction) compared to the classical production routes
State of play: The AO process consumes organic solvents (2 – 5 kg per ton of H_2O_2) and quinone (0.5 - 2.5 kg per ton of H_2O_2)[8], and has incomplete recovery or regeneration of chemicals. Furthermore, it requires noble metal catalysts (Pd), which cannot be always completely recycled: Losses of 0.1-0.3 g per ton of H_2O_2 correspond to 120-360 kg Pd per year. Additionally, H_2O_2 producers generate emissions of about 10 g volatile organic compounds (VOC) and 15 m^3 waste water with 0.6 kg total organic carbon (TOC) per ton H_2O_2 produced. These water streams require further treatment. **Project contribution:** The POWER2HYPE process does not require organic solvents and does not produce any critical waste. Moreover, the utilization of noble-metal-free materials for cathodes rather than palladium will have an impact on the EU competitiveness, as this critical raw material's supply chain comes from unstable markets such as Russia. Furthermore, environmental burdens through toxic wastes associated with the mining of precious metals are avoided. **Main beneficiaries/stakeholders:** i) H_2O_2 producers being able to offer sustainable and low-waste production technologies; ii) H_2O_2 end-users with ambitions regarding sustainable supply chains; iii) civil and political society through a new technology contributing to an environmentally friendly economy.
KPI (associated): 0% use of organic solvents, noble metal catalysts or other critical materials.
Contribution to KSO: C and A **Contribution to expected impact in cluster 4:** Circular economy **Contribution to specific objectives of the Processes4Planet:** S.O. 3 Moving towards commercially viable climate neutral and circular industry solutions
Outcome 6: Competitive costs of the new process technology and its integration in the processing line, including upstream and downstream
State of play: AO process plants comprise infrastructure for H_2 supply (could be replaced by H_2 electrolyzers), chemical reactors for AQ hydrogenation and H_2O_2 synthesis including the required safety infrastructure, extraction columns, distillation columns for purification and concentration. Operation costs of the AO process are driven by energy costs, by required replacement of the solvents (incomplete recovery), and by catalyst replacement. **Project contribution:** CAPEX and OPEX of the POWER2HYPE process will be based on the specific application scenario (e.g. plant size, local electricity costs). The required system components are limited to the electrolysis stage, possibly a purification step (if H_2O_2 contains traces of salts, acids or bases), and a concentration stage. The relatively simple set-up results in low manufacturing costs. Operation costs are dominated by electricity costs. Here, the POWER2HYPE approach of co-producing H_2O_2 in the anodic AND cathodic half-cell lead to doubled faradaic efficiency compared to a single anodic or single cathodic process, thus halving operation costs. Additionally, full marketability of the product can be expected due to its 100% "renewability" bonus under CO_2 emission trading systems, such as the EU's ETS, potentially compensating for slightly higher cost of production, when compared to the conventional production pathway. Coupling with the energy sector through "electrification" of chemical production will help creating a stable and balanced renewable energy landscape, with reduced grid violations and hence reduced energy costs for the entire society. **Main beneficiaries/stakeholders:** i) H_2O_2 producers; ii) H_2O_2 end-users interested in supply with economically competitive sustainable H_2O_2 or in local (on-site) and on-demand H_2O_2 production for own use.
KPI (associated): Economic competitiveness of the POWER2HYPE technology confirmed by techno-economic assessment, realistically assuming current density of 400 mA/cm^2, FE of 160% or more (over both half-cells) and specific energy consumption of 3 kWh/kg_{H2O2} or less for the electrosynthesis step.
Contribution to KSO: C, A and D **Contribution to expected impact in cluster 4:** new business models **Contribution to specific objectives of the Processes4Planet:** S.O. 3 Moving towards commercially viable climate neutral and circular industry solutions

In addition to the impacts and outcomes abovementioned, POWER2HYPE contributes to the following Sustainable Development Goals (SDG):

- **SDG 6: Clean water and sanitation:** POWER2HYPE technology allows also small-scale and remote H_2O_2 production, where the product can be used for drinking water treatment, sanitation and disinfection. It is planned to validate the applicability of the H_2O_2 produced in Task 6.3 in industrial processes, in collaboration with industrial end-users as external partners. As part of this validation, it will be tried to also show the applicability in water treatment, sanitation or disinfection, thus validating the potential contribution of the POWER2HYPE contribution to SDG 6.
- **SDG 7: Affordable and clean energy:** H_2O_2 is not a useful energy carrier, but electrochemical production contributes to effective coupling of energy sector and chemical industry, thus enabling buffering of load peaks and balancing renewable energy supply over time. Contribution of POWER2HYPE to this SDG to be validated through examination of the load flexibility of the process in Task 6.3.
- **SDG 12: Responsible consumption and production.** POWER2HYPE technology represents a totally renewable production pathway, based on the consumption of renewable energy, water and oxygen from air as only raw materials. Problematic waste streams and emissions of hazardous compounds are avoided. It is therefore a technology option for responsible production. Contribution of POWER2HYPE to SDG 12 can be validated at the end of the project through evaluation of the list of raw materials used for the developed process with respect to their toxicity and through evaluation of the KPI for Outcome 5 (0% use of organic solvents, noble metal catalysts or other critical materials).
- **SDG 13: Climate action.** POWER2HYPE production process is expected to be completely CO_2-neutral. Contribution of POWER2HYPE to SDG 13 can be validated based on the results of Task 7.3 (LCA) and through corresponding evaluation of the KPI with respect to Outcome 3 (reduced specific CO_2 emissions by 90% or more).

2.1.2. Framework conditions, regulations, and policies

The European Green Deal

In line with the **European Green Deal**, POWER2HYPE shall be a positive force in the European chemical industry's ambition to become climate neutral by 2050. Reaching this goal will only be possible with the help of climate-neutral and circular economy solutions developed by the industrial players themselves, in our case, the development and implementation of an electricity-based process to produce H_2O_2 that can benefit from a progressive decarbonization of the power sector.

Moreover, the project will contribute to the **EU Chemicals Strategy for Sustainability**, which is the first step towards a zero-pollution ambition for a toxic-free environment announced in the European Green Deal. In line with the strategy's aim to significantly increase the protection of human health and the environment from harmful chemicals, POWER2HYPE will develop a flexible technology that will produce H_2O_2 on-site at nearly any concentration. This feature will remove the need to handle, transport and store highly concentrated H_2O_2 solutions thus increasing substantially the safety of the process. No organic solvents are required in the **POWER2HYPE** production process and no critical waste streams are generated. All materials (including catalysts) are free of noble metals, thus avoiding upstream wastes, e.g. in mining processes.

Moreover, the impact of a **greener production of hydrogen peroxide on its end-markets** will be substantial. As already introduced on Figure 10, H_2O_2 is utilized in several industries with a large carbon footprint that will be decreased thanks to POWER2HYPE. The environmental burden of conventionally produced H_2O_2 that is currently propagated to the downstream processes where H_2O_2 is used, will be drastically reduced by shifting H_2O_2 production to POWER2HYPE technologies.

Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH)

Along with the EU Chemicals Strategy for Sustainability, the European Commission has begun to work on a revision of the REACH regulation in January 2022. It is expected, that the electrochemically in POWER2HYPE produced H_2O_2 can be considered as "drop-in" product, so that no new REACH registration will be necessary. However, if there are any impurities or, e.g., electrolyte salts from the anodic half-cell reaction present in the H_2O_2 solutions, a new registration might be required.

Regulations and specifications related to specific applications

The requirements in terms of quality and purity of H_2O_2 depend on the specific on the targeted application. As the generated H_2O_2 is expected to be very pure and of high quality, no severe implications regarding to its application are expected. However, the quality and suitability of produced H_2O_2 will be validated as part of WP6 (Demonstration).

2.1.3. Barriers

The following potential barriers for the industrial implementation of POWER2HYPE technology have been identified:

- **Availability of cheap carbon-neutral energy** that represents a very substantial part of, or could even exceed, the low-carbon power capacities anticipated by the International Energy Agency (IEA) in 2050. This a general challenge for all Power-to-X technologies, also in the context of growing e-mobility and of other sectors transitioning to renewable energy base. This challenge can only be met through intensified scale-up of renewable electricity generation, particularly in regions with favorable solar and wind potentials. This scale-up must be a top priority of policies in the EU and in all member states.
- **Higher production costs related to using low-carbon technologies**. This is also a general challenge for renewable alternative production pathways in comparison with conventional fossil-based processes. Competitiveness will increase through rising prices of fossil feedstocks (such as currently induced by the Russian war in Ukraine) and common price volatility in fossil markets. Competitiveness will also increase through enhanced efficiency from technical learning curves, once the POWER2HYPE technology will be industrially implemented. Furthermore, regulatory frameworks, such as the EU ETS, provide an economic advantage for sustainable production pathways. Finally (but importantly), price premiums for "green" products are to a certain extent accepted by users who advertise the sustainability of their products.
- **Investment requirement**. Large-scale roll-out of new production processes require substantial investments. The main barrier in this context is the risk and uncertainty related to such investments. Companies need to have the certainty that there is a stable market for new product, in particular in the light of a cost premium of newly introduced renewable products (compared to conventional fossil-based counterparts). Planning reliability must be provided through stable political and regulatory frameworks, clearly and reliably favoring truly sustainable production processes.
- **Manufacturing experience and prove of concept on a large industrial scale is missing**. Key technologies for POWER2HYPE, such as the specific electrochemical cell, electrodes, membranes and the concentration unit, have not been developed and proven at scale. Even though electrochemical process and reactors in general have already been industrially implemented (e.g. in chlor alkali and water electrolysis), manufacturing of specific POWER2HYPE components has still to be established. To do so (up to pilot-scale) is one of the project's objective. Beyond POWER2HYPE, more R&D efforts should be dedicated to the development of gas-diffusion electrodes (GDE) and related technologies (membranes, electrochemical cells, etc.), as such technologies are crucial for successful implementation of electrochemical conversion of gaseous educts. More dedicated R&D projects will help to readily provide the technologies when they are needed at relevant scale. An important gap to be filled also is to build a solid knowledge base on the investments cost to be expected when scaling the process further up. Techno-economic studies on the relation between CAPEX, OPEX, process performance, market and regulatory boundary conditions are crucial to better understand financial requirements and risks associated to industrial rollout of the POWER2HYPE technology and other electrochemical processes. For this reason, there is an entire work package dedicated to technology impact assessment (WP7) in POWER2HYPE. Such assessment studies should generally be part of all R&D projects on renewable technologies, processes and products.
- **Social acceptance of sustainable chemicals and awareness creation.** The social acceptance of the POWER2HYPE technology itself is not expected to represent a barrier, as it offers the possibility to replace a largescale chemical production process that entails environmental risks and GHG emissions by a sustainable carbonneutral process generating now harmful waste streams. However, the POWER2HYPE approach requires substantial input of renewable electric energy. Consequently social acceptance of the scale-up of solar and wind power plants in conjunction with the POWER2HYPE approach are addressed in a sociotechnical analysis to provide policy recommendations for a successful roll-out of the technology.

2.2. Measures to maximize impact

The actions to be done to properly maximize POWER2HYPE impacts are classified as follow:

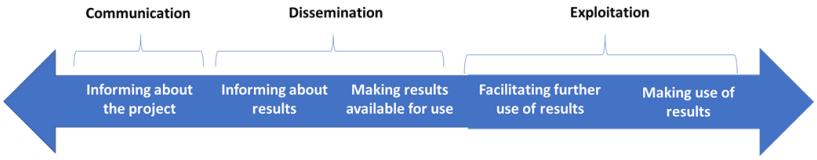

Figure 13: Areas of activities to maximize the impact of POWER2HYPE.

This section describes the communication, dissemination, and exploitation activities to maximise the impact of the POWER2HYPE project.

2.2.1 POWER2HYPE External Advisory Board (EAB)

To consider the key end-user requirements and facilitate the further market uptake of the innovative POWER2HYPE process, early in the project (M6), the project partners will set-up an External Advisory Board, mostly composed of industrial end-users to be involved during the project (see section on exploitation below for more details). The EAB will provide technical support to the project and supervise and validate the accomplishment of objectives. This has a technical intention, but also a commercial and dissemination goal. EAB members will be regularly (at least once per year) join part of the regular consortium meetings where key project findings will be presented and discussed. Advisory Board members give feedback and recommendations to adjust the focus of R&D activities, in agreement with the work plan.

2.2.2. Dissemination and exploitation of results

2.2.2.1. Dissemination plan

As initial activity, relevant external stakeholders and the targeted audience for the POWER2HYPE dissemination activities are identified. The appropriate channels here selected will be fine-tuned during the project and the identified tools will be adapted to the project's case. The following dissemination plan describes the potential of the POWER2HYPE consortium for outreach at a regional/national and EU levels by support and use of the free-ofcharge services of the EU-funded world's largest support network for SME, Enterprise Europe Network (EEN). The key dissemination goals are: i) To make POWER2HYPE results, findings and technologies known to relevant scientific communities, relevant industrial stakeholders and interested public; to enhance reach of the project at local, national and European and international level; to pave the way for exploitation of POWER2HYPE results, findings and technologies.

<u>Key stakeholders and main target audience</u>

1. **Industries utilizing hydrogen peroxide:** This includes various sectors such as other chemical industries, pulp & paper, mining activities, textile industries, food processing & packaging, wastewater treatment processes, electronics, etc. POWER2HYPE main industrial stakeholders and end-users will be approached through associations' events and also: Participation in scientific and technology/industry conferences introducing new technology developments and the new technology proposed

    a. Participation in scientific and technology/industry conferences;
    b. Publications in industry-relevant and scientific journals;
    c. Participation in fairs related to technologies and final product applications.
2. **Associations in the chemical sector and end markets:** members and decision-makers of the following associations will be approached:
    a. **CEFIC:** the European Chemical Industry Council is the main European trade association for the chemical industry.
    b. **EuChemS:** the European Chemical Society is an umbrella organization representing national Chemical Societies and other chemistry-related organizations in Europe.
    c. **Cepi:** European association representing the paper industry (with four standing committees, taking longterm strategic perspectives on issues affecting the industry, including environment and safety.
    d. **Euromines:** representative of the European metals and minerals mining industry
3. **Academic and research institutions** (including universities and R&D centers, Ph.D. candidates, postdocs, etc.): Scientific activities within a collaborative space where formal and informal teams and networks promote sharing best practices and experiences on electrochemistry and sustainable processes. Synergies, cooperations and vivid

exchange with other R&D projects thematically related to POWER2HYPE will facilitate advances in the development in the field of electrochemical production processes beyond the state of art.

4. **European agencies and platforms active in sustainable processes:** The project will be in line with the Processes4Planet partnership's objectives. POWER2HYPE results will be disseminated through participation in key events (conferences, brokerage events etc.), encouragement of knowledge sharing in the platforms' communities and communication through partners already members of such platforms. The following initiatives will be:
    a. **Processes4Planet** (SPIRE members IRIS, FHG and Solvay):
    b. **ECHA**: the European Chemicals Agency
    c. **EIT Manufacturing**: Innovation Community within the European Institute of Innovation & Technology (EIT) – that connects the leading manufacturing actors in Europe
    d. Relevant agencies and departments of the **European Commission:** including DG CLIMA, DG CONNECT, DG ENER, CINEA, EISMEA, REA, DG RTD, etc.
5. **Regulatory/national authorities:** The POWER2HYPE pilot plant will be located in one of Solvay's sites in Belgium. Therefore, regional authorities and the general public in Belgium will be interviewed through surveys to study issues related to social acceptance and socio-economic impact potentials of the POWER2HYPE technologies. Additionally, publications in the general press and pilot visits will be used to raise public awareness, interest and acceptance.
6. **Third parties/joint ventures:** Relevant stakeholders, including technology developers, engineering services, chemical producers, etc., will be approached in fairs and other events with the objective of involving them in POWER2HYPE events and meetings (WP8) and of initiating joint exploitation activities.
7. **General public/civil society:** The project POWER2HYPE will disseminate and communicate its results to the general public by popularizing the information in national and local media as well as in social media including screen casts and videos.

Dissemination channels and related communication actions and tools

There are several possible channels for disseminating information and project results. The selection of modalities and ways of channeling information and project image may vary in relation to the communication targets. The different modalities and targeted audience (initial and non-exclusive selection) that will be used by the POWER2HYPE consortium are depicted in Table 7 below.

Table 7: Dissemination and communication actions and tools.

Description	Related KPI(s)	Audience	Target(s)
Topic discussions	# discussions	All	500
Meetings	# meetings	All	50
Pilot visits at Solvay's premises (location of the POWER2HYPE pilot plant)	# visits # participants # feedback	1, 3, 4, 5, 6	5 10 5
Events: Conferences, fairs, info sessions (e.g on EU-funded conference like EURONANOFORUM and IndTECH, SusCHEM....)	# events # contacts made	All	10 50
Open access publications	# references	1, 3, 4, 6	500
R&D seminars/webinars i.e., EU Green Week 2024	# attendees # feedback	3	10
Organization of mid-term and final project symposium	# participants	1, 2, 3, 4, 6	20
Summer school, Austrian Summer School of Energy Transition ASSET 2024	# PhD, Master students	1, 3	1-60
Popular science articles/ interviews in national and local media	# articles	All	15
Progress reports	# reports	EC	4
Presentations	# presentations # attendees	All	10-20 20-50
Policy recommendations	#	1, 2, 4, 5	1
Website	website statistics	All	1
Social media channels (Twitter, Facebook, Research gate, etc.)	# followers	All	100

Scientific papers	# scientific papers	3	14
Logo and Templates	# produced	All	1
Screencasts and Videos	# produced	All	8

The partners will also mention POWER2HYPE on their own website and will, on the project website and their own social media channels, release the projects main results to the targeted audience.

- **FHG** will use their own website (www.igb.fraunhofer.de/en) as well as LinkedIn (http://www.linkedin.com/company/fraunhofer-igb), Twitter (https://twitter.com/FraunhoferIGB), Instagram (www.instagram.com/fraunhoferigb) and other channels for dissemination and communication of results. FHG will also provide own scientific publications via their own Open Access repository.
- **TRSA** will use their own website (www.tecnicasreunidas.es) and LinkedIn and Twitter channels to disseminate results: https://www.linkedin.com/company/tecnicas-reunidas/ and https://mobile.twitter.com/tecnicas_reun @Tecnicas_reun.
- **SGL, DIA, IRIS, Solvay** will use as a main online channel their websites: https://www.solvay.com/en/ ; www.sigracet.com. www.diaccon.de. https://www.iris-eng.com/.
- **EI** will list **POWER2HYPE**, include a short description and post news/events of the project in their website: https://energieinstitut-linz.at/. They will also use their account on social media and platforms to disseminate key results of the project, namely:
    - LinkedIn: https://www.linkedin.com/company/energieinstitut-an-der-jku-linz/mycompany/.
    - Twitter: https://twitter.com/energie_inst.
    - Peer-reviewed papers in ResearchGate and various scientific platforms (Google Scholar, Zenodo etc.): https://www.researchgate.net/institution/Energieinstitut_an_der_Johannes_Kepler_Universitaet_Linz -

WU and **WR** will announce the project on its website (www.wur.eu) and their LinkedIn channel.

- **WU** will use example of the project (after clearance) as case in education
- **SLVG** will announce the results on their website (www.solvge.com) as well as on LinkedIn.

Pre-defined conferences, fairs and workshops

For events out of Europe, the coordinator will verify the eligibility before authorizing the assistance of the partners.

Table 8: Selected conferences with planned attendance by POWER2HYPE partners.

Partner(s)	Conferences	Date
FHG, WR	European Symposium on electrochemical engineering	Annually
FHG, WR	Annual Meeting of the International Society of Electrochemistry (ISE)	Annually in summer
EI	“Green & Sustainable Energy Conference”, Elsevier	Annually in summer
EI	“Conference on Sustainable Development of Energy, Water and Environment Systems » (SDEWES). In Germany, Croatia and other EU countries (alternating)	Annually in October/November
EI	“World Sustainable Energy Days” (WSED), in Wels/Austria:	Annually in spring
EI	“Internationale Energiewirtschaftstagung” (IEWT), in Graz/Austria:	Annually in spring
EI	“ECEEE Summer Studies”, different locations in Europe:	Annually in summer
EI	Ökobilanzwerkstatt, DACH countries	Annually, in September
WU	North American Catalysis meeting	Biannually
WU	Europacat	Biannually
WU, WR	Dutch Chemisty and Catalysis Conference NCCC	Annually
DIA, FHG	ACHEMA Fair	2025
General	IndTECH, EURONANOFORUM	Annually
General	SusCHEM	Annually

Planned peer-reviewed publications

Gold/green open access will be granted for the publications and related data.

*Table 9: Publications planned by scientific **POWER2HYPE** partners.*

Partner	Publications
EI	Three Open Access articles on a) Techo-economic & market perspectives, b) Life Cycle Assessment, c) Overall impact assessment of POWER2HYPE technology
FHG	Three Open Access articles on process development of anodic H_2O_2 generation, testing novel electrode materials and integration of anodic and cathodic half-cell reaction
WR	At least two open access articles on a) ORR of H_2O_2 at a cathodic MEA, b) production of concentrated H_2O_2 by paired electrolysis
WU	At least three open access articles in journals like journal of catalysis, applied catalysis B, and a PhD thesis
FAU	At least three open access articles on the development of new BDD-based electrodes in journals like Energy Materials or Diamond rel. Mat.; 1 PhD thesis

Articles will also be submitted to channels of EC, such as CORDIS Wire, and HORIZON RESULTS PLATFORM. In addition to scientific papers, the partners will also place publications in industry journals and magazines to inform potential end-users of the POWER2HYPE applications, and to raise awareness of the general public for its application and benefits.

2.2.1.2. Exploitation plan

In the following, the exploitation strategy and benefits per partner are depicted, followed by an outline of the project's business cases that will be further developed during the project.

Individual exploitation plan

Table 10: Exploitation plan per partner

Partner	Result	Benefits and strategy for exploitation
FHG	**Electrochemical WOR process development and integration**	As the world's leading applied research organization, FHG acts as a link between basic research and industrial application, supporting innovation processes by developing novel and new technologies to a maturity level, from where it can be further progressed by industrial partners towards commercialization. In POWER2HYPE, FHG will develop the anodic H_2O_2 production process and will contribute to the electrochemical cell design and process integration. This

		development is highly IP-relevant, and Associated with document Ref. Ares(2022)7691108/11/2022FHG plans to protect the generated IP through patents, where possible, exploit the IP through i) further own development in the course of future publ funded projects, ii) collaborative development with industrial partners tow commercial application and iii) through licensing. Furthermore, FHG will dissemi results through scientific publications, such as articles and conference contribution well as through exhibitions / fairs. This will also lead to increasing FHG's reputa and acquiring further publicly funded projects as well as collaborations with indus partners.
WR	**Electrochemical ORR process development and integration**	As one of Europe's largest applied research organizations in agricultural and life sciences, WR acts as a link between fundamental research and industrial application, supporting innovation by developing novel technologies to advanced lab or pilot scale, from where it can be transferred to industrial partners for commercialization. In POWER2HYPE, WR will develop the cathodic H_2O_2 production process and will contribute to electrochemical cell design and process integration. This development is highly IP-relevant, and WR plans where possible, to protect potential new generated IP through patent applications and will exploit the IP through i) further own development in the course of future publicly funded projects, ii) collaborative development with industrial partners towards commercial application and iii) through licensing. Furthermore, WR will disseminate results through scientific publications, such as manuscripts and conference contributions, as well as through exhibitions and fairs. The dissemination activities will also allow increasing WR's reputation and acquiring further publicly funded projects and collaborations with industrial partners
WU	**Design of efficient and stable carbon-based ORR catalyst**	The mission of WU BCT is to develop catalytic conversion strategies relevant for a sustainable future based on fundamental scientific insights. Electrocatalysis is key to achieve sustainable conversions when renewable electricity is used. Within POWER2HYPE we will develop new carbon based catalysts for the production of hydrogen peroxide from air/oxygen and water using electrocatalysis. The approach of the group is based on developing fundamental knowledge but with an eye to the application. The knowledge gained in POWER2HYPE on new catalysts and integrating the catalyst with other components of a H_2O_2 production cell e.g. gas diffusion layers and membranes will be shared with the other partners and published in scientific papers, and conference contributions after protecting it by IP (when relevant). IP will be transferred to one of the industrial partners when appropriate and desired, after negotiating proper conditions (WU does not commercialize IP itself). This approach will help WU to keep and expand its scientific and applied profile, which will help in acquiring new projects to contribute further to the mission of the group.
FAU	**Stable, efficient and low-cost BDD electrodes**	The development of efficient, stable and low-cost boron doped diamond electrodes enables a new wide research area for fundamentally scientific and applicationdriven projects. FAU hopes to increase the scientific skills of personnel and students related to BDD-electrodes. This should help the industry to introduce new electrochemical processes, which replace CO_2 generating techniques.

DIA	**Electrochemical stack**	DIA is one of a few producers in the world of diamond electrodes in an industrial scale having 20 years of experience in the technology, as well as in coating technology. Thanks to POWER2HYPE, DIA will enable the transfer of the results regarding electrode manufacturing to production and commercialization. They will also guide partners in their development regarding feasibility in industrial electrode production. After the POWER2HYPE project, DIA will be able to offer a new portfolio of customized electrolyzers using GDEs, membranes and new types of diamond electrodes. The main customer will be manufacturers using electrochemical synthesis processes.
EI	**Sustainability assessment**	EI will oversee all sustainability topics in POWER2HYPE, including the modelling of the upscaling scenarios, the technoeconomic assessment (TEA), LCA and overall impact assessment (including stakeholder/societal acceptance, and ecological aspects). POWER2HYPE will allow EI to gain insight into the
		H_2O_2 markets and the latest technologies in the sector.Associated with document Ares(2022)7691164 - 08/11/2022 EI will use the project results for refined analyses subsequent projects, i.e., also to identify further research. Also, some of the publis results are planned to add to the dissertation projects of involved EI researchers. KPIs quantified in WP 7 help with targeted upscale and marketing of POWER2HY technology development.
Solvay Group	**H_2O_2 producer Membrane technologies**	As a key player in the chemical sector, Solvay is one of the main stakeholders in POWER2HYPE and bring their view as a H_2O_2 manufacturer. As such, SOLVAY main aim is to explore new production technologies where virtually only water and electricity is needed and, therefore, with a substantially decreased environmental footprint. Furthermore, in POWER2HYPE, Solvay will design and develop new ionomers for membrane and electrodes for more robust electrochemical and fuel cells devices.
TRSA	**Pilot scale integrated electrolyser**	TRSA is committed with the transition to zero-carbon energy supply by closely working with its stakeholders to achieve the IEA's Sustainable Development Scenario (SDS). TRSA corporate target is to get increasingly involved in projects with a clear positive impact related to decarbonization and do it as fast and as fair as possible. In line with this, TRSA is collaborating in several initiatives for the escalation of electrolysis technologies and other industrial process electrification. POWER2HYPE increases the portfolio of those initiatives by contributing to the development of a novel electrochemistry-based technology for the industrial production of H_2O_2. TRSA acts as technology developers that will exploit the prototype as privative equipment for H_2O_2 production. TRSA may also make the electrolyser as part of a technology package for the implementation of a H_2O_2 production plant, including (pre-)feasibility studies, basic and detail engineering, and eventual EPC.
SGL	**Gas diffusion electrodes**	SGL is one of the world leading manufacturers of carbon-fiber based materials for fuel cells, flow batteries and electrolysers. SGL's vision is to explore new markets for carbon fiber-based materials in view of expanding their current portfolio and find new customers in new markets. More specifically to POWER2HYPE, they will learn about the critical design criteria of gas diffusion electrodes for H_2O_2 production and develop new product types based on the experience gained throughout the project. Their direct client profile will be electrode/stack/system manufacturers that utilize large quantities of hydrogen peroxide.

SLVG	**DSP for H_2O_2 Concentration**	As a pioneer in the field of concentration of H_2O_2, SolvGE strives to make H_2O_2 accessible to any company at any concentration, enabling several initiatives to transition to green oxidizer chemicals. SolvGE´s technology is able to concentrate H_2O_2 from low concentrations (3-10%) to any desired concentration up to 98%. SolvGE's goals are in line with this project as it ensures high concentration of H_2O_2 can be utilized in a variety of industries. SolvGE wants to ensure that electrochemically produced H_2O_2 becomes accessible to all industries by processing the raw feedstock into any required product quality.
IRIS	**Electrochemical process digital twin**	IRIS is a technology developer, offering solutions to make end-users optimize their processes, making them more sustainable and reducing their carbon footprint. In the frame of POWER2HYPE, IRIS will develop new capabilities and competences on Digital Twins to control and monitor chemical processes (such as electrochemical conversion), which work with renewable electricity. This project will allow them to increase their client portfolio in the chemical sector. IRIS will be able to offer monitoring solutions, data acquisition and Digital Twins for process optimization to industries willing to switch from chemical to electrochemical production of H_2O_2.

Business case for the POWER2HYPE electrochemical process

In a nutshell, the initial POWER2HYPE business case will follow Solvay's current strategy to sell decentralized small-scale units for hydrogen peroxide production based on the AO process[47], with the technology developers of the POWER2HYPE acting as key partners.

Table 11: POWER2HYPE initial business case.

<table>
<tr><td rowspan="2">Key partners
Solvay Group
TRSA
DIA
SGL
IRIS</td><td>Key activities
Know-how in H_2O_2 production
Manufacturing & integration of decentralized production units</td><td rowspan="2">Value proposition
Sustainable product
Flexibility of the process (upstream & downstream)
CO_2-free
Safety
Independent of unstable markets
Decreased timeline of installation due to digital twin</td><td>Customer relationships
Long-lasting expertise in H_2O_2 production and electrolysis Training of staff</td><td rowspan="2">Customer segments
Pulp & Paper
Propylene oxide
Aquaculture
Aseptic Packaging
Chemicals
Cleaning in Place
Etchant/Photovoltaic
Electronics
Gas scrubbing
Healthcare
Homecare
Mining
Personal Care
Soil treatment
Textiles
Wastewater Treatment</td></tr>
<tr><td>Key resources
Proprietary technologies & know-how on the market
Exploitation partnership</td><td>Channels
Solvay's portfolio of clients and worldwide presence
Other channels coming from experienced partners</td></tr>
<tr><td colspan="3">Cost structure
Engineering procurement personnel, integration, testing and commissioning Cost of components
Infrastructure</td><td colspan="2">Revenue streams Sale of assets</td></tr>
</table>

The market for POWER2HYPE[48]

Besides the individual exploitation strategies explained above for each partner, the POWER2HYPE exploitation depends on the market for hydrogen peroxide as commodity. Its market was valued at 5,187.32 kiloton in 2020, and is projected to register a CAGR of over 3% during the forecast period (2021-2026).

[47] https://www.solvay.com/sites/g/files/srpend221/files/tridion/documents/myH2O2_Brochure_LR%20%281%29.pdf
[48] https://www.mordorintelligence.com/industry-reports/hydrogen-peroxide-market/

Focus on Europe

In general, the market was negatively impacted by COVID-19 in 2020, however, the currently rising demand from the pulp and paper industry and increasing consumption of hydrogen peroxide as raw material for propylene oxide production (HPPO Process) are expected to drive the market growth. In 2018, the global market volume for hydrogen peroxide in the pulp and paper industry was 2,200.93 kiloton, and is estimated to reach about 2,820 kilotons by 2024, with a CAGR of about 5%, which will stimulate the demand for hydrogen peroxide in the coming years and make it one of the key markets for POWER2HYPE. More specifically, the European pulp and paper industry had a production capacity of 43.65 million ton in 2019, with a growth rate of 1.49% compared to the previous year, which, in turn, stimulates the market demand for hydrogen peroxide.

In view of this trend and in line with the POWER2HYPE implementation roadmap (Figure 11), exploitation and implementation activities will focus on these sectors and applications. Assuming that POWER2HYPE technology is applied to cover the newly generated capacity (5% CAGR) from the point in time when it has reached commercial scale (2030), **the market share of this new technology can be projected to reach 40% of the total European capacity by 2040** (4.2 Mio tons/year).

A global perspective

Newer opportunities for hydrogen peroxide in wastewater treatment and the growth of the electronic industry in developing Asian countries are likely to create opportunities for the market in the coming years. Furthermore, the Asia-Pacific region is expected to dominate the market and is also likely to witness the highest CAGR during the forecast period.

For instance, China is the largest pulp and paper producing nation in the world, producing about 125 million ton of cardboard and paper in 2019, with a growth rate of about 4% compared to the previous year, which, in turn, stimulates the regional market demand for hydrogen peroxide. The electronics industry in China is also the largest in the world. Due to these factors, the market for hydrogen peroxide in the region is expected to have a steady growth during the forecast period.

By 2050, counting on having growth still in the European market (taking market share up to 60%), with new adopters, and assuming a conservative market penetration worldwide (taking over less than the new capacity created, 3% Associated with document Ref. Ares(2022)7691164 - 08/11/2022

CAGR), **we believe that** POWER2HYPE **technology could nearly displace the AO process, with over 80% of the capacity installed using the project's technology**.

Competitors

The hydrogen peroxide market is consolidated in nature, with the top five players occupying nearly 70% of the market. The majority of companies are focusing on expanding their production capacities, primarily in the AsiaPacific region. Some of the major players in the market include Solvay (key industrial partner in POWER2HYPE), Evonik Industries AG, Mitsubishi Gas Chemical Company INC, Arkema, and Kemira,, among others.

POWER2HYPE SWOT analysis

*Table 12: **POWER2HYPE** SWOT Analysis*

STRENGTHS	WEAKNESSES
Partners' increasing knowledge and developments in new markets and initial customer portfolio is extensive Sustainable product Flexibility of the process (upstream & downstream) CO_2-neutral Increased safety compared to the SoA process Independent of unstable feedstock supply (e.g. NG) Decreased timeline of installation due to digital twin	Electrosynthesis step alone does not reach the same concentrations levels as the AO process (yet): Up to 10% H_2O_2 compared to up to 40% H_2O_2 Actual overall performance (electrochemical reactions and DSP (concentration)) uncertain and yet to be proven as one of the major outcomes of the project technoeconomic viability to be validated
OPPORTUNITIES	**THREATS**

EU strategy pushes to move to electrochemical process The current 2050 targets in climate matters need best cases to finance Current geopolitical situation advantageous for the approach as availability and prices of fossil resources are uncertain and volatile; this will trigger "sustainable" investment decisions Labeling of sustainable products are attractive	Availability of cheap low-carbon energy may not increase as expected R&D investments are needed and investors my avoid risks associated with less mature technologies Social acceptance of chemicals and awareness creation for sustainable produce require extra efforts

2.2.1.2. Post project plan

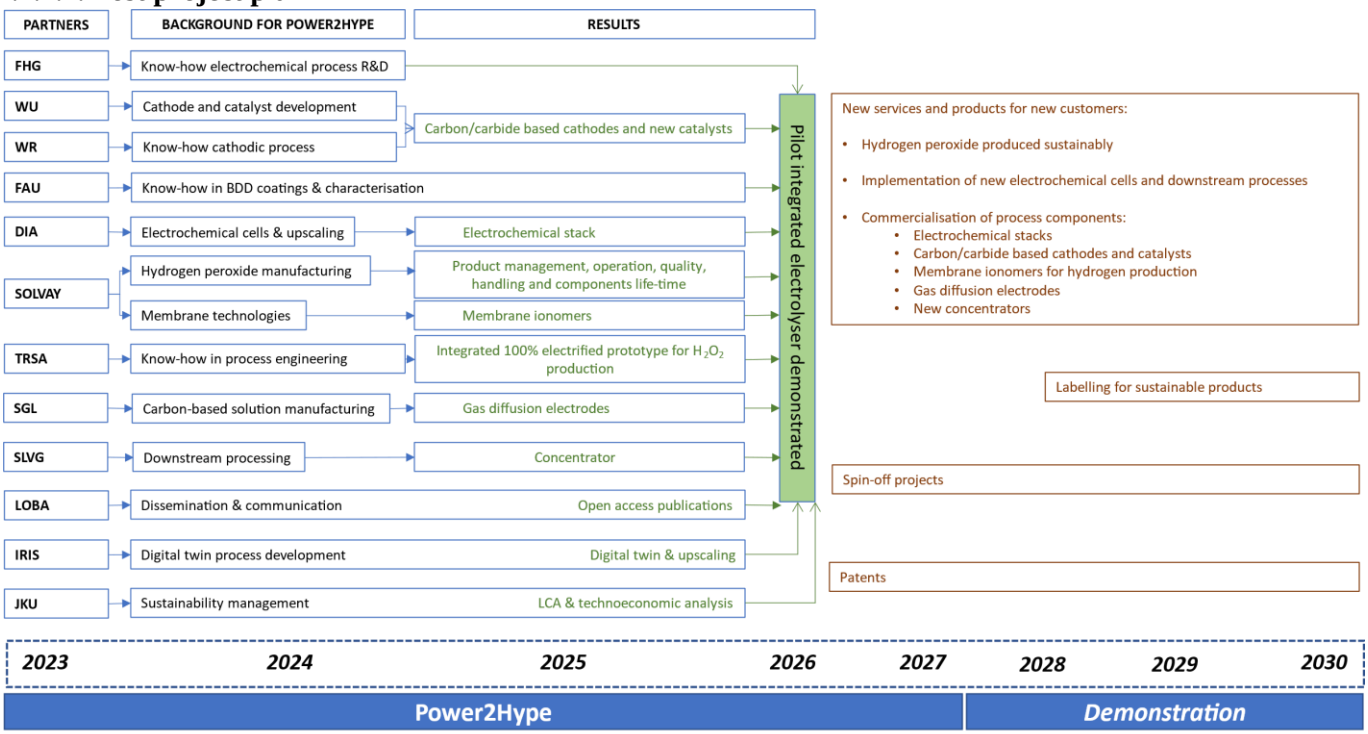

Figure 14: POWER2HYPE Post-project plan

2.2.2. Communication plan

In addition to the plan for dissemination of scientific and technical results, a full communication strategy is introduced (WP8). A corresponding communication plan will be developed and implemented in the course of the project (including with the dissemination plan), also ensuring that has continuation beyond the end of the project's duration. The key communication channels will be the following and will be highly used during key dissemination activities:

- **Website:** The POWER2HYPE project will provide several communication and community tools and will include a website via a web publishing component. The webpage will be constantly maintained and updated to communicate the project progress.
- **Social Media.** Social media campaign will be designed to integrate Online Media (Facebook, Twitter, LinkedIn, YouTube), Paid Media (the vision will be led internally but paid services will be used for marketing automation, among others) and Shared Media. We will adopt a "highly effective impression" strategy, a strongly segmented target by using the different channels for specific targets/objectives.
    - A detailed 'Social Media Content Strategy' will be defined at the beginning of the project defining specific strategies for each social media.
    - Facebook and YouTube will be used to present the results popularizing the information to be understood by the general public.
    - LinkedIn: to influence whole market and attract decision makers.
    - Twitter will be used mainly to alert the industrial sector, in direct relation with the project's activities. ○ Research Gate will be used to promote the project in the scientific community

- **Events:** The large Community Event Programme will also attract the public. The events will provide sessions (e.g., keynotes, open discussions), which invites European citizens to participate and to discuss with experts in electrochemistry and sustainable processes.
- **Scientific publications:** in peer-reviewed journals and presentations in relative thematic conferences and seminars.
- **Popular science publications:** in national- and local media as well as in online popular science periodical, or university newspapers.

2.3. Summary

Table 13: Impact Summary

SPECIFIC NEEDS	EXPECTED RESULTS	D&E&C
Currently H_2O_2 production relies on non-sustainable process (energy consumption and related CO_2 emissions, use of noble metal catalysts, use of organic solvents and related VOC emissions, generation of hazardous wastes). It is also mainly produced in large centralize chemical units, while market would require more decentralized units delivering lower volume.	At the end of the project it is expected that the consortium will have developed a process and its key components for an innovative electrochemical route to produce H_2O_2. The innovative process will be demonstrated at pilot scale in a relevant environment at Solvay's premises (TRL6) with the following key characteristics: i) carbon-neutral (essentially zero CO_2 emissions); ii) no use of organic solvents and critical (rare) materials, only based on carbon catalysts, water, oxygen (air) and renewable electric energy; iii) suitable for decentralized small-scale units and on-demand production; iv) capable of meeting requirements of a broad range of applications, even in niche markets requiring very low (1-3%) or very high (>70%) H_2O_2 concentration.	The membership of Solvay as main EU H_2O_2 producer (>50% of EU market) will serve the purpose of disseminating and exploiting the achieved results. The dissemination of project results to the relevant community in the EU will be performed via technical conferences and publications in scientific journal as well as technical expert groups. The creation of an external Advisory Board to be composed mainly of industrial stakeholders (such as H_2O_2 end-users) will support market uptake of the process. The partners are very active in sharing knowledge and practical experience in the scientific community and in the H_2O_2 sector.

TARGET GROUPS	OUTCOMES	IMPACTS
At the end of the project the electrochemical H_2O_2 process will be at TRL6, so industrial development involving consortium members will be required to bring it to TRL9 (by 2035). As expressed in section 2.2, the project outcomes are expected to be commercially exploited primarily by key industrial partners of the POWER2HYPE project that will work in collaboration to mature the new	The main outcome of the POWER2HYPE project will be to achieve a **sustainable, decentralized, efficient, and flexible electrochemical process to produce H_2O_2**. This process is expected to be economically viable and the project will identify the key end-user applications that should be targeted after 2030.	1. switch from chemo-catalytic process to an electrochemical process for H_2O_2 production 2. a highly sustainable process leading to remarkably reduced CO_2 emissions and zero emission of hazardous waste stream. 3. sustainable components without noble metal or critical materials and catalysts 4. no use of organic solvents 5. economically viable renewable energy-based process

process. Solvay will play a key role by proposing the POWER2HYPE to **key H_2O_2 end-users for various applications (key target group)**. Other key target groups will be regulatory/ national / EU authorities to move regulation to push for such sustainable electrochemical processes.	POWER2HYPE will also lead to findings on more sustainable components for electrochemical stacks that could be used for other cutting-edge applications (fuel cells, electrosynthesis, etc).	Associated with document Ref. Ares(2022)7691164 08/11/202 After starting commercialization from 2030, POWER2HYPE technology has the potential to supply about 40% of the **EU H_2O_2 market (4.2 Mio tons/year** leading to significant CO_2 emissions **savings of approximately 6.7 Mio tons CO_2/year**)

3. Quality and efficiency of the implementation

3.1 Work plan and resources

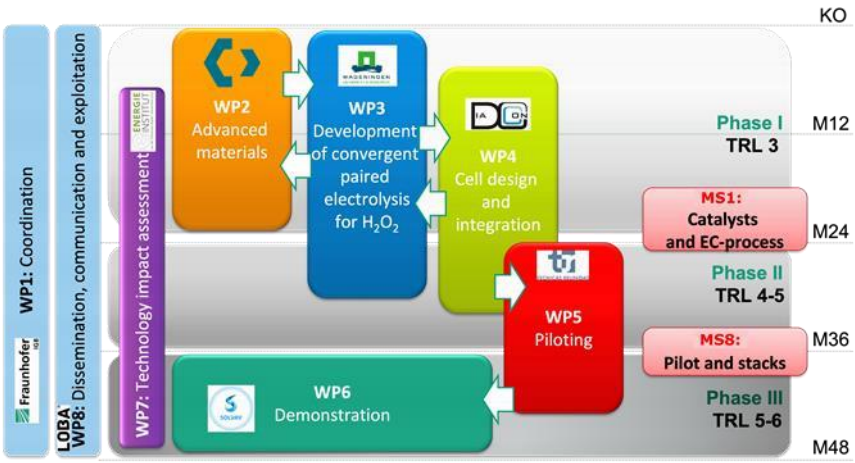

Figure 15: POWER2HYPE structure of the work packages with decision milestones and progressive TRL. The timing depicts the main activities of the WPs. KO = kick off

This section contains the project's work plan and it is thus the core of this proposal. POWER2HYPE is divided into eight work packages in line with the project's approach (see Section 1.3). The project work is divided into 8 work packages (WP) that will be completed over three distinct project time phases 1 to 3, with two milestone decision points between them (Figure 15). All activities related to project management and coordination are bundled in WP1. R&D activities (WP2-WP6), LCA and techno-economic analysis (WP7) and exploitation actions, dissemination & communication activities. Each WP produces a set of deliverable, has to achieve critical milestones and follows a well-defined and appropriate timeline, all described in the present work plan.

WP2 is dedicated to the development of advanced and industrially viable materials and cell components for the electrochemical process and reactor systems. Specifically, materials-related research in WP2 is devoted to catalysts, gas diffusion layers/electrodes, ion-exchange membranes and BDD anodes.

WP3 deals with the development and an efficient coupling of anodic and cathodic reactions to achieve an optimal convergent paired electrolysis for H_2O_2 production. **WP4** aims at the design of an electrochemical reactor paired H_2O_2 production and at the development of the downstream processing for further H_2O_2 concentration.

WP5 will focus on the development, building, commissioning and testing of a pilot plant to produce H_2O_2 using air, water and electricity. This pilot plant will be explored in **WP6** to demonstrate the entire POWER2HYPE process for producing H_2O_2 under industrially relevant operational conditions, with a focus on technical feasibility, stability, load-flexibility, cost-efficiency and scaling factors, always in view of a future commercialisation. WP6 also includes tests to validate the application of the produced H_2O_2 in industrial processes (such as bleaching or epoxidation processes), in collaboration with associated industrial partners (H_2O_2 users).

WP7 is dedicated to technology impact assessment, including prospective lifecycle assessment (LCA), technoeconomic analysis (TEA), examination of social acceptance and model-based (in silico) design of a scaled-up production system.

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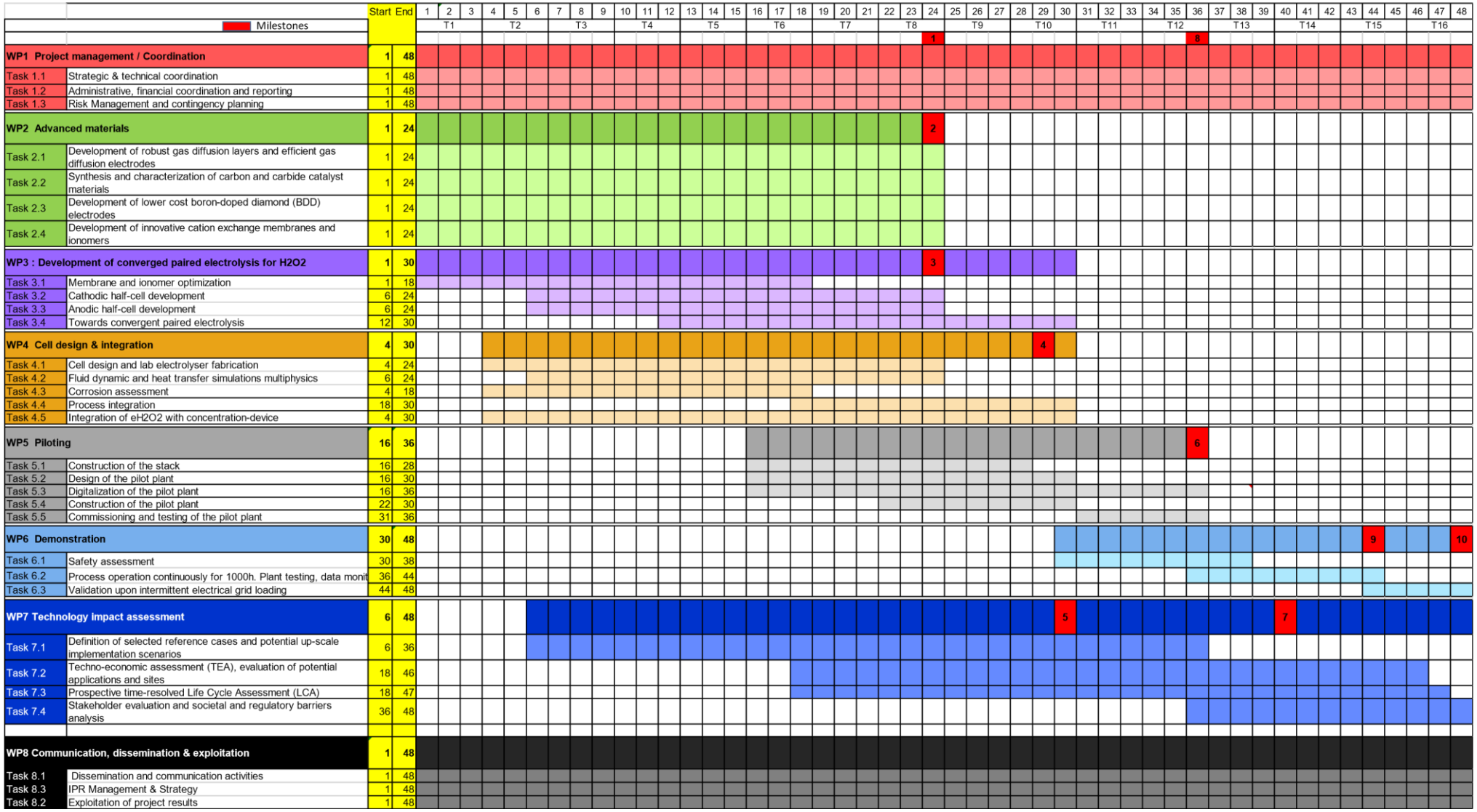
Milestones

		Start	End
WP1	**Project management / Coordination**	**1**	**48**
Task 1.1	Strategic & technical coordination	1	48
Task 1.2	Administrative, financial coordination and reporting	1	48
Task 1.3	Risk Management and contingency planning	1	48
WP2	**Advanced materials**	**1**	**24**
Task 2.1	Development of robust gas diffusion layers and efficient gas diffusion electrodes	1	24
Task 2.2	Synthesis and characterization of carbon and carbide catalyst materials	1	24
Task 2.3	Development of lower cost boron-doped diamond (BDD) electrodes	1	24
Task 2.4	Development of innovative cation exchange membranes and ionomers	1	24
WP3	**: Development of converged paired electrolysis for H2O2**	**1**	**30**
Task 3.1	Membrane and ionomer optimization	1	18
Task 3.2	Cathodic half-cell development	6	24
Task 3.3	Anodic half-cell development	6	24
Task 3.4	Towards convergent paired electrolysis	12	30
WP4	**Cell design & integration**	**4**	**30**
Task 4.1	Cell design and lab electrolyser fabrication	4	24
Task 4.2	Fluid dynamic and heat transfer simulations multiphysics	6	24
Task 4.3	Corrosion assessment	4	18
Task 4.4	Process integration	18	30
Task 4.5	Integration of eH2O2 with concentration-device	4	30
WP5	**Piloting**	**16**	**36**
Task 5.1	Construction of the stack	16	28
Task 5.2	Design of the pilot plant	16	30
Task 5.3	Digitalization of the pilot plant	16	36
Task 5.4	Construction of the pilot plant	22	30
Task 5.5	Commissioning and testing of the pilot plant	31	36
WP6	**Demonstration**	**30**	**48**
Task 6.1	Safety assessment	30	38
Task 6.2	Process operation continuously for 1000h. Plant testing, data monit	36	44
Task 6.3	Validation upon intermittent electrical grid loading	44	48
WP7	**Technology Impact assessment**	**6**	**48**
Task 7.1	Definition of selected reference cases and potential up-scale implementation scenarios	6	36
Task 7.2	Techno-economic assessment (TEA), evaluation of potential applications and sites	18	46
Task 7.3	Prospective time-resolved Life Cycle Assessment (LCA)	18	47
Task 7.4	Stakeholder evaluation and societal and regulatory barriers analysis	36	48
WP8	**Communication, dissemination & exploitation**	**1**	**48**
Task 8.1	Dissemination and communication activities	1	48
Task 8.3	IPR Management & Strategy	1	48
Task 8.2	Exploitation of project results	1	48



3.2 Capacity of participants and consortium as a whole

The consortium of POWER2HYPE consists of 15 legal entities located in Germany, Belgium, the Netherlands, Austria, Italy, Spain, Portugal, & France. These partners cover the whole value chain of the H_2O_2 production including key components development and production up to construction and operation of H_2O_2 plants. The complementarity of partners' skills will ensure the upraise of POWER2HYPE technology from TRL3 up to TRL6 within the project timeline. Key industrial partners will lead the further steps towards industrialization and commercialization beyond the end of the project.

The consortium is composed of **two Universities** (Wageningen University, **WU** and Friedrich-Alexander-Universität Erlangen-Nürnberg, **FAU**) and **three research institutions** (Fraunhofer Society, **FHG**; Wageningen Food & Biobased Research, **WR**; Energieinstitut an der Johannes Kepler Universität Linz, **EI**). **FHG** has vast experience coordinating European projects, communicating with EU project officers and an excellent record of accomplishment in anodic H_2O_2 production. **WR** has remarkable experience in ORR. Together, FHG and WR will pave the way for efficiently pair H_2O_2 electrosynthesis. **EI** will communicate with all WPs to monitor and contribute to the sustainability and economic viability of the process. **FAU** and **WU** have outstanding experience in carbon-based materials synthesis and characterization, thereby setting the ground for a sustainable electrocatalytic process. POWER2HYPE has **ten companies on board, 6 industries, 3 SMEs and 1 startup**. Among the companies are four entities of the industrial **Solvay group** (Solvay SA, **SOLVAY**; Rhodia Operations, **RHDO**; Solvay Specialty Polymers Italy SPA, **SSPI**; Rhodia Laboratoire du Futur, **SRLF**), bringing in expertise in both membrane development and H_2O_2 production. The SME DiaCCon (**DIA**) is a specialized developer of electrolyser and industrial-scale BDD electrodes. The company SGL Fuel Cell Components GmbH (**SGL**) is one of the world's leading manufacturers of carbon components as gas diffusion layers and bipolar plates. The SME Iris Technology Solutions (**IRIS**) and the industry Tecnicas Reunidas (**TRSA**) support the consortium with strong skills in engineering and digitalization (**IRIS**) as well as chemical plant engineering and piloting (**TRSA**). The innovative start-up SolvGE (**SLVG**) is developing H_2O_2 concentrator units for downstream processing of raw and diluted H_2O_2 streams. Finally, the SME GLOBAZ (**LOBA**) provides extensive experience in dissemination and communication activities. More details are shown below in Table 17 on each partner's key expertise and role within the project.

Table 17: Partner key expertise and equipment infrastructure

Partner	Key expertise (WP	Key equipment / Infrastructure related)
FHG	Collaborative project coordination (WP1); Anodic process (WP2, 3,4)	Electrochemical equipment (potentiostats, RRDEs, screening systems with 6 parallel cells, flow cells, pumps) and capacity for TRL4-5 experiments. Analytics (GC, HPLC, NMR, etc.). Materials characterization (SEM, XRD, etc.).
WU	Cathode Materials (WP2,3)	Electrochemical equipment (Potentiostats, etc.), analytical facilities (e.g. SEM-EDX, XPS, XRD)
WR	Cathodic ORR towards H_2O_2 (WP2,3)	Electrochemical equipment, catalyst preparation equipment (rotary bed , catalyst calcination and reduction reactors) characterization equipment(STEM-HAADF, XPS, XRD, ICP-OES, SEM-EDX, (ATR) FTIR, surface area and porosity analyzer)
FAU	CVD coating with diamond and metal nitrides/borides/carbides, PVD metal or metal carbide coatings (WP2,3)	5 CVD diamond coating machines, 1 CVD TiNB coating machine, 1 PVD coating machine, electrochemical long-time testing, material characterization (dilatometer, SEM, microprobe, ion mill etc.)
DIA	Diamond coating and electro-chemical stack production (WP4,5)	4 diamond coating machines, coating thickness measurement tools
EI	LCA, TEA, new value chains assessments (WP7)	Licence of lifecycle software GaBi® ts 9.5 (LCA) SuperPro Designer® for assessment of production processes An in-house tool CoLLeCT for TEA

		Toolbox for qualitative socio-technical assessments
Solvay (Group)	Membrane development (WP2,3) H_2O_2 production (WP5,6,7)	SoA R&D facilities for ionomers development and testing. SoA R&D facilities for H_2O_2 development, quality and testing
TRSA	Engineering of pilot plants (WP4,5,6,7)	Lab analysis equipment for chemical characterization Rooms and infrastructure dedicated to construction of pilot plants
SGL	Gas diffusion electrodes (WP2)	GDL testing, electrochemical labs, & GDL manufacturing lines
SLVG	Patented H_2O_2 concentrator technology (WP4, 5,6)	R&D facility for development and scale-up of H_2O_2 concentrator units Hardware for interfacing H_2O_2 production and concentration
LOBA	Communication, dissemination. & exploitation activities (WP8)	Software to develop and provide various communication tools
IRIS	Process modelling and monitoring., data mining (WP5)	Required software to develop the digital twin of the POWER2HYPE pilot plant.

4. Ethics and Security

4.1 Ethics

The Ethics Opinion on the Power2Hype proposal in the Ethics Summary Report sets Ethics clearance (the proposal is 'ethics ready').

The beneficiaries must ensure that all ethics issues related to activities in the grant are addressed in compliance with ethical principles, the applicable international and national law, and the provisions set out in the Grant Agreement. This includes the ethics issues identified in the Ethics Summary Report (if any) and any additional ethics issues that may emerge in the course of the grant. In case any substantial new ethics issues arise, beneficiaries should inform the granting authority.

*Table 18: Ethics Issues Table*Associated with document Ref. Ares(2022)7691164 - 08/11/2022

Section	Yes/ No	Comment
1. Human Embryonic Stem Cells and Human Embryos		
Does this activity involve Human Embryonic Stem Cells (hESCs)?	No	
Does this activity involve the use of human embryos?	No	
2. Humans		
Does this activity involve human participants?	No	
Does this activity involve interventions (physical also including imaging technology, behavioural treatments, etc.) on the study participants?	No	

Does this activity involve conducting a clinical study as defined by the Clinical Trial Regulation (EU 536/2014)? (using pharmaceuticals, biologicals, radiopharmaceuticals, or advanced therapy medicinal products)	No	
3. Human Cells / Tissues (not covered by section 1)		
Does this activity involve the use of human cells or tissues?	No	
4. Personal Data		
Does this activity involve processing of personal data?	Yes	The proposed activities include the implementation of interviews and surveys from regional authorities and general public in Belgium to study issues related to social acceptance and socioeconomic impact potentials of the generated technologies. However, we confirm that no personal data will be collected.
Does this activity involve further processing of previously collected personal data (including use of preexisting data sets or sources, merging existing data sets)?	No	
Is it planned to export personal data from the EU to non-EU countries? Specify the type of personal data and countries involved	No	
Is it planned to import personal data from non-EU countries into the EU or from a non-EU country to another non-EU country? Specify the type of personal data and countries involved	No	
Does this activity involve the processing of personal data related to criminal convictions or offences?	No	
5. Animals		
Does this activity involve animals?	No	
6. Non-EU Countries		
Will some of the activities be carried out in non-EU countries?	No	
In case non-EU countries are involved, do the activities undertaken in these countries raise potential ethics issues?	No	
It is planned to use local resources (e.g. animal and/or human tissue samples, genetic material, live animals, human remains, materials of historical value, endangered fauna or flora samples, etc.)?	No	
Is it planned to import any material (other than data) from non-EU countries into the EU or from a non-EU country to another non-EU country? For data imports, see section 4.	No	
Is it planned to export any material (other than data) from the EU to non-EU countries? For data exports, see section 4.	No	
Does this activity involve low and/or lower middle income countries, (if yes, detail the benefitsharing actions planned in the self-assessment)	No	
Could the situation in the country put the individuals taking part in the activity at risk?	No	
7. Environment, Health and Safety		

Does this activity involve the use of substances or processes that may cause harm to the environment, to animals or plants.(during the implementation of the activity or further to the use of the results, as a possible impact) ?	Yes	The work carried out at lab scale (gram scale) is safe. The waste generated is disposed according to State legislation. We follow all the safety measures according to the rules of the university/research institute as well as MSDS (Material Safety Data Sheet) of a chemical provided by the supplier. And the samples that are generated are fully tracked in internal management systems, containing MSDS (Material Safety Data Sheet), including for samples resulting from running the prototype.
Does this activity deal with endangered fauna and/or flora / protected areas?	No	
Does this activity involve the use of substances or processes that may cause harm to humans, including those performing the activity.(during the implementation of the activity or further to the use of the results, as a possible impact) ?	Yes	We follow state legislation for the proper disposal of chemical waste, thus avoiding any harm to humans.
8. Artificial Intelligence		
Does this activity involve the development, deployment and/or use of Artificial Intelligence? (if yes, detail in the self-assessment whether that could raise ethical concerns related to human rights and values and detail how this will be addressed).	Yes	AI tools are purely technical and do not relate to activities affecting human rights and values.
9. Other Ethics Issues		
Are there any other ethics issues that should be taken into consideration?	No	

Associated with document Ref. Ares(2022)7691164 - 08/11/2022HE